



C H A P T E R 10

Electronic Payment Systems

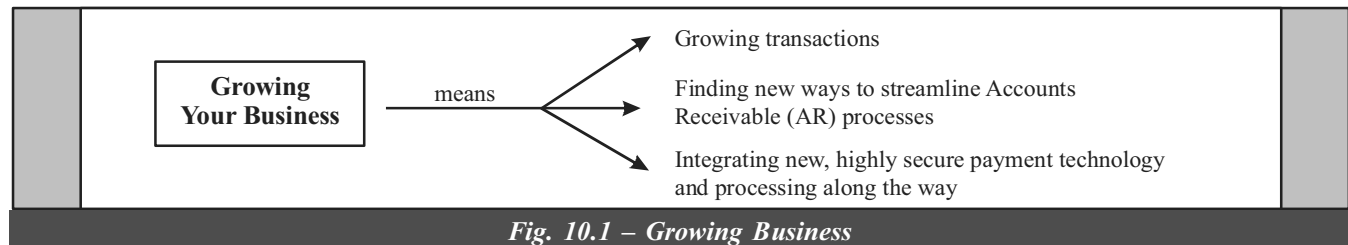
- | | |
|--|--|
| <i>10.1 Introduction</i> | <i>10.15 Net Banking</i> |
| <i>10.2 Types of Electronic Payment System</i> | <i>10.16 Enabling Regulations</i> |
| <i>10.3 Payment Method Types</i> | <i>10.17 Critical Building Block for Electronic Commerce</i> |
| <i>10.4 Receipt of Payments</i> | <i>10.18 The Offline and Online Purchasing Processes</i> |
| <i>10.5 The Traditional Payment System</i> | <i>10.19 The Shopping Process with Payseal</i> |
| <i>10.6 Electronic Funds Transfer (EFT) Systems</i> | <i>10.20 ICICI Plan to Create a Bank for Virtual Customers</i> |
| <i>10.7 Modern Payment System</i> | <i>10.21 Paperless Cheque Processing</i> |
| <i>10.8 Electronic Cash</i> | <i>10.22 E-Cash</i> |
| <i>10.9 The Steps for Electronic Payment</i> | <i>10.23 Wallets Have Made Easy for Consumers</i> |
| <i>10.10 Payment Security</i> | <i>10.24 Miscellaneous Topics:</i> |
| <i>10.11 Information Sources for Electronic Commerce and Payments on the Web</i> | <i>– Dynamic Pricing</i> |
| <i>10.12 Internet Protocol Security (IPSec)</i> | <i>– Redefining Financial Freedom (Countrywise) (Alluva)</i> |
| <i>10.13 Traditional Payment Methods</i> | |
| <i>10.14 Problems with Traditional Payment Methods</i> | |

10.1 INTRODUCTION

An e-payment system is a way of making transactions or paying for goods and services through an electric medium, without the use of cheques or cash. It is also called **electronic payment system** or **online payment system**.

Increase customer convenience with trusted online payment processing solutions – Adopt simplified acceptance of credit cards ACH, EBPP and more (EBPP is Electronic Bill Presentment and Payment).

An estimated 56% of organisations still use siloed EBPP and online payment systems, resulting in \$ 1 billion in added operations cost annually.

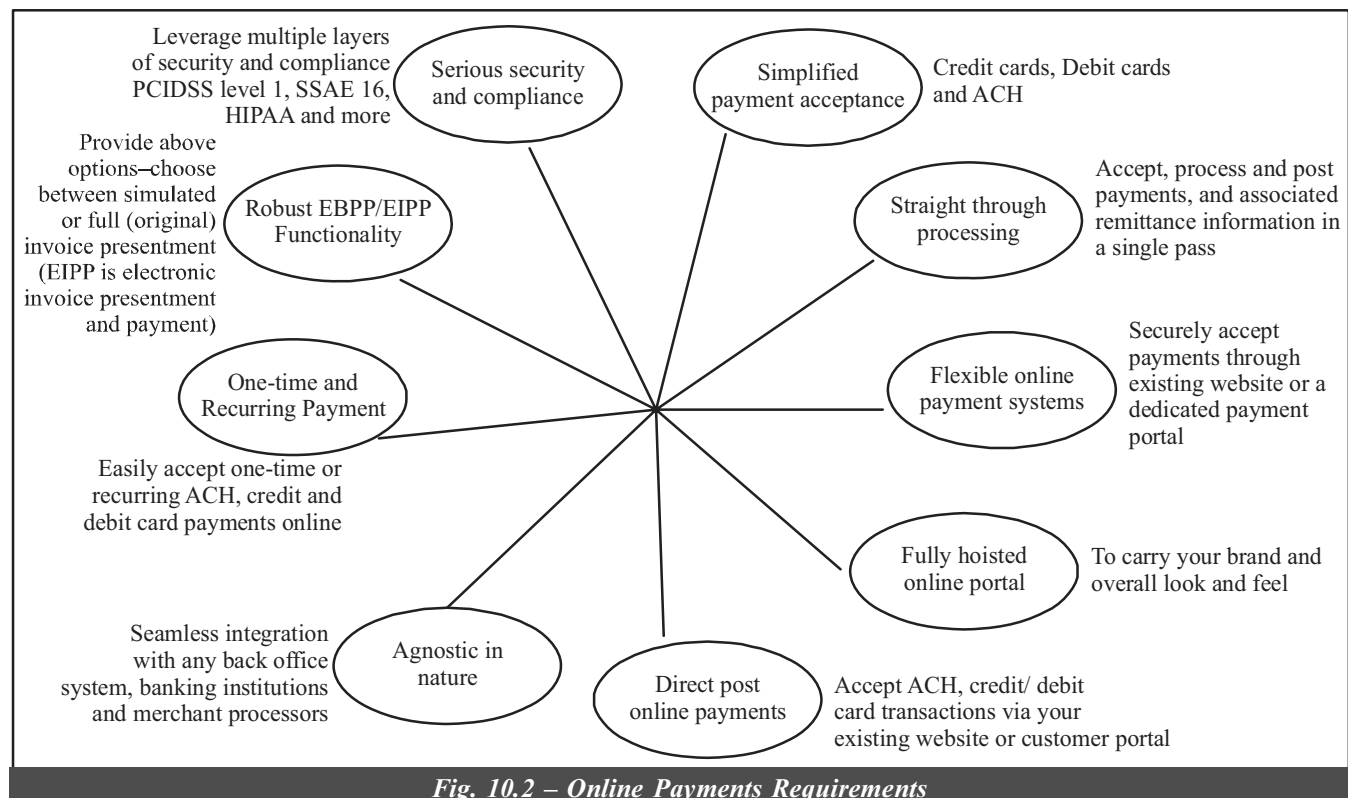


A strategic mix of benefits to our national processing:

- A fully compliant consolidated receivables platform.
- Ability to automate traditionally manual, time-consuming processes.
- Ability to scale and easily configure electronic transactions to meet specific processing needs across our national business made for an excellent fit.

The online payments evolution should be there in the organisation built in for customers with the control and oversight needed by leveraging a seamlessly scalable SaaS (Software as a Service) platform and the industry leading security, compliance and truly integrated platform that only a good platform can deliver.

Online paying system should have the following:



This is for customer expectation for:

- Speed, ease of use, and variety of choices (plenty of payment options)
- Types of online payments available (methods)
Amazon (Google/e-Bay/Apple payments, etc.

The Cost of Inaction

The cost of using siloed receivables and payment processing systems (100%)

results in

- Decreased customer adoption of fragmented electronic options – 53%
- Lost productivity from siloed payment system – 32%
- In long call times due to the use of disparate payment acceptance systems in call centers – 15%

Electronic Bill Presentation and Payment (EBPP) Service works as under:

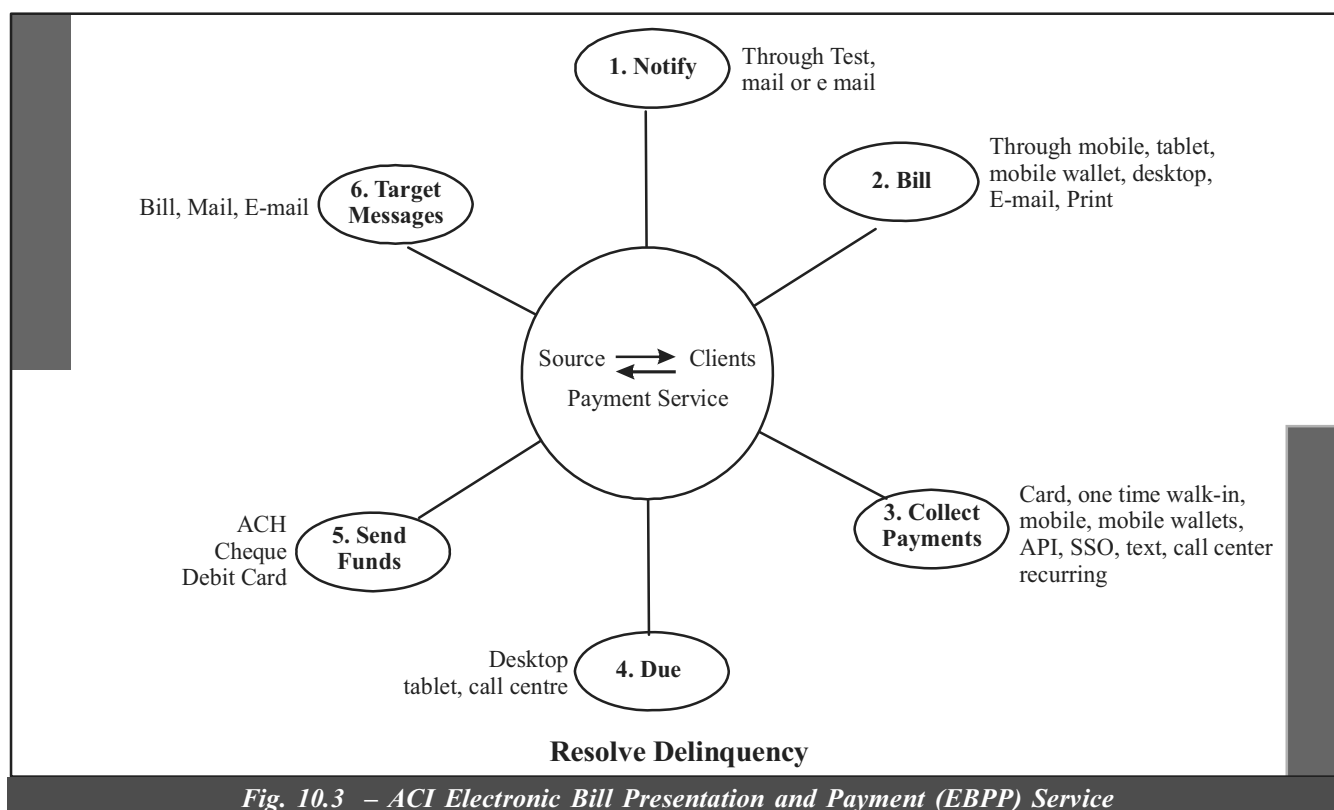


Fig. 10.3 – ACI Electronic Bill Presentation and Payment (EBPP) Service

Electronic Payments first emerged with the development of wire transfers. Electronic Fund Transfer (EFT) is an electronic transfer of information that equates to moving funds from one financial institution to another. Electronic payment systems are alternative cash or credit payment methods using various electronic technologies to pay for products and services in electronic commerce. The process is complicated because of the near anonymous electronic nature of transactions taking place between the networked computer systems of buyers and sellers. Here, many security issues are involved. The e-commerce payment process is also complex because of the wide variety of debit and credit alternatives and financial institutions and intermediaries that may be part of the process. Hence, a variety of electronic payment systems have evolved. Electronic cheque is a type of electronic funds transfer that can be given to an online company for deposit at an online financial institution.

10.2 TYPES OF ELECTRONIC PAYMENT SYSTEM

At least dozens of electronic payment systems proposed or already in practice are found. The groupings can be made into three broad categories on the basis of what information is being transferred online.

Electronic funds transfer is the secure transfer of funds via an agreed protocol and system. In the recent days, banking and finance operations have long been dependent on EDI and EFT to ensure that money and securities are transferred to the company or individual that requested the action. Engineering firms manufacturing products have used business-to-business project management, manufacturing and outsourcing for the development and delivery of their products. Nowadays, people are afraid to operate through the Internet using their credit card information. Your information can be used against

you. Because of this, one wonders whether any business could be done at all. Customers are far more wary about providing credit information than ever before.

The greatest threat to future Internet sales growth is the press story waiting in the wings about some company that has had its customer base hacked for credit card numbers, names, expiration dates, home addresses, e-mail addresses and even products recently purchased.

To reduce the threat of this potential disaster, a number of new companies are trying some innovative approaches. Companies that pay the vendor through the Internet without sending your personal credit information or banks that use electronic cheques are examples of new services that are reducing the risk of providing credit information.

10.3 PAYMENT METHOD TYPES

These are of two types which involves flow of funds inside corporations:

- Payments received from your customers and
- Payments made to your suppliers (including employees for the labour and governments that supply bureaucracy).

E-commerce helps in reducing these expenses by streamlining the cash collection and payment operations. While automating the supply chain, the receipt and payment processing functions could be automated along with the rest of the supply chain. The receipt of an actual product or confirmation of delivery to the customer could automatically be deposited when they are received through the use of credit card software.

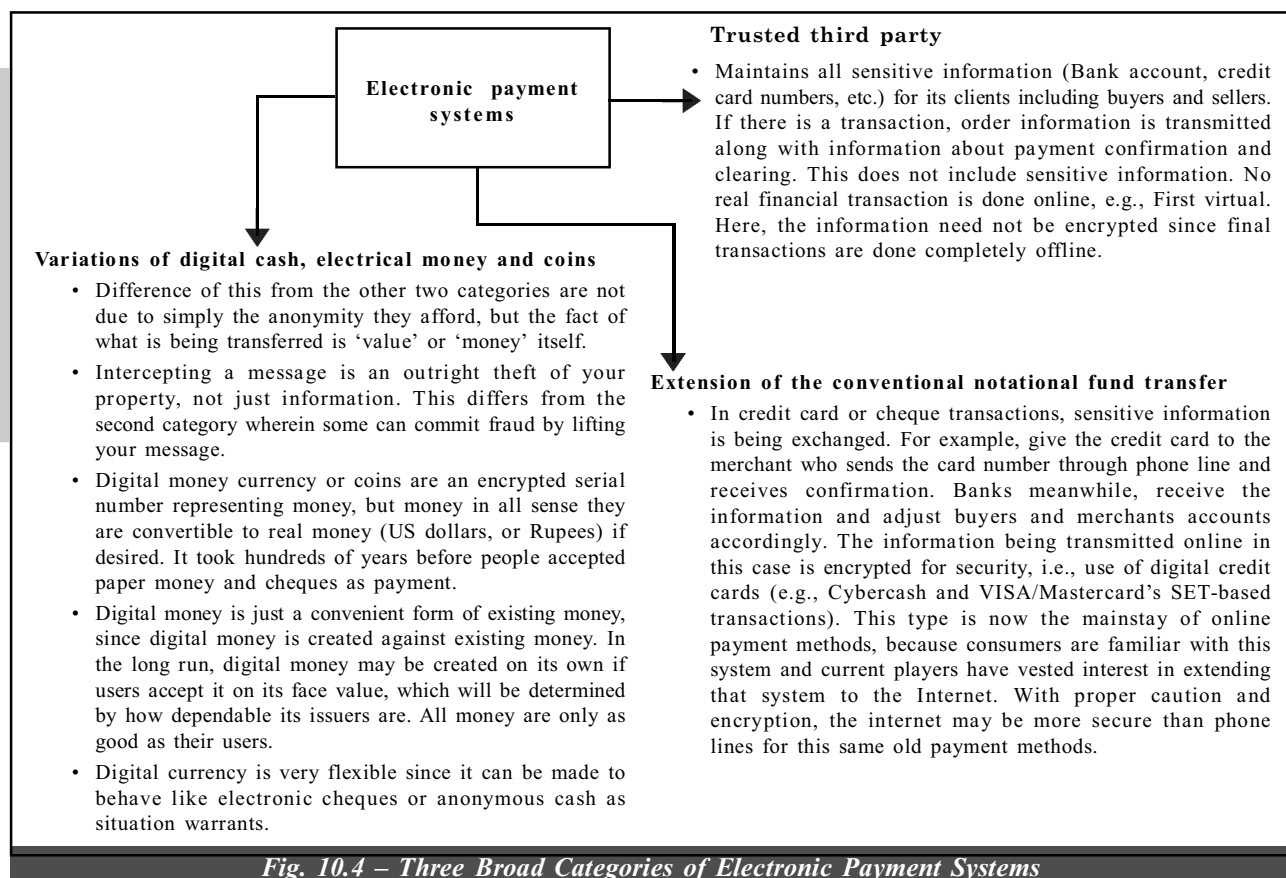


Fig. 10.4 – Three Broad Categories of Electronic Payment Systems

Payment Method Types (Options: Cash, Cheques, Credit/Debit Cards, Mobile Payments, Electronic Bank Transfers)

Following are the methods for payment in general:

- | | | |
|---|-------------------|-----------------------|
| (i) Credit cards: As global payment solution, credit cards are the most common way for customers to pay online. | | |
| (ii) Mobile Payments | (v) Prepaid cards | (viii) Cheque |
| (iii) Bank Transfers | (vi) Direct Debit | (ix) VISA Master card |
| (iv) E-Wallets | (vii) Cash | |

The most common alternative payment methods are:

- | | | |
|---------------------|------------------------|----------------------|
| (x) Debit cards | (ix) Money orders | (xvi) Mobile Payment |
| (xi) Change cards | (xiii) i-Phone payment | (xvii) Amazon Pay |
| (xii) Prepaid cards | (xiv) Smart Cards | |
| (xi) Direct debit | (xv) Net Banking | |

Credit cards, cheque, wire transfer and cash are acceptable in US.

The ten excellent online payment systems are popular:

- | | |
|------------------------------------|--------------|
| • Authorise Net (Most widely used) | • Dwolla |
| • PayPal | • Stripe |
| • Google Checkout | • Brain Tree |
| • Amazon | • Samurai |

Payment methods in international trade used around the world today include:

- | | |
|---------------------------|---------------------|
| • Cash in advance | • Letters of credit |
| • Documentary collections | |

Cards used in banks and businesses (global e-business also):

- | | |
|--------------------------|-----------------------|
| • VISA and Master card | • Bank Transfer |
| • Maestro cards | • JCB cards |
| • Paypal | • Union Pay (China) |
| • American Express cards | • Direct Debit |
| • Diners Club cards | • Sofort Banking etc. |

E-commerce payment methods:

- | | | |
|---------------------------------|----------------------|-----------------------------|
| • Union Pay (Chinese Bank card) | • Amazon Pay | • Payment Request API (W3C) |
| • Google Pay | • e-Bay | • E-Wallet |
| • Apple Pay | • Google Pay | • Pay Pal |
| • Gift Cards | • Digital Currencies | |

Payment Methods in India – For Diverse Range of Cultures (4.5 billion People in Asia)

Electronic payment and settlement systems in India:

- | | |
|--|---|
| • Electronic clearing service (ECS credit) | • Credit and Debit cards (ECS Debit) |
| • Real-time gross settlement | • National Electronic Funds Transfer (NEFT) |
| • India-Nepal Renaissance Facility Scheme | • Immediate Payment Service (IMPS) |

Digital Payment Method:

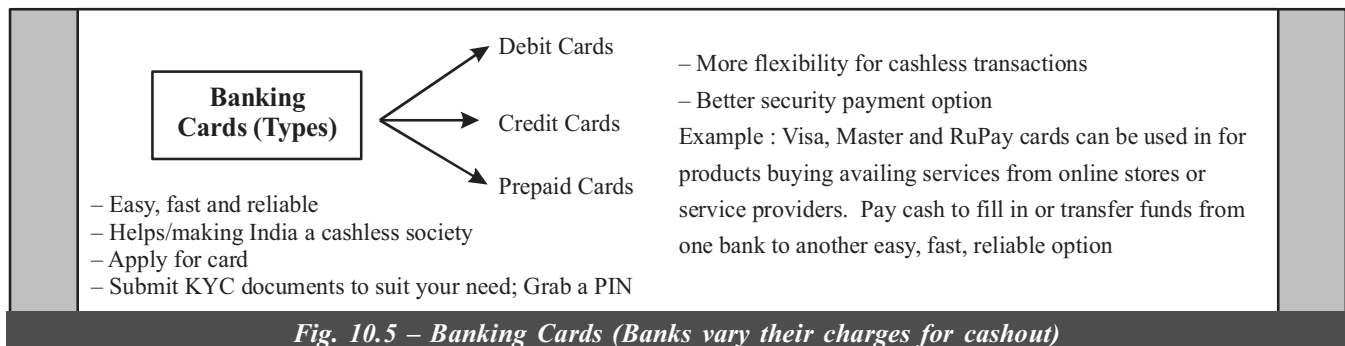
- Banking cards (Debit/Credit) provides more security, control and convenience
- USSD (Unstructured Supplementary Service Data) for Mobile Transaction
- AEPS (Aadhar Enabled Payment System)
- UPI (Unified Payment Interface)
- Mobile Wallets
- Banks Prepaid Cards
- Points of Sale (POS)
- Internet Banking
- Amazon Pay

Amazon Pay (Another convenient, secure and quick way to pay. Use your information which is already stored in Amazon account credentials to login and pay at leading merchants' websites and apps (where you love to shop or home delivery).

Become a Seller Today

Put your products in front of the crores of customers on Amazon everyday. One of the latest ways of making **online payment** are through mobile phones, instead of credit card or cash. The customer has to do is to send a payment request for his/her service provider via text message, The customer's mobile account is charged for the purchase.

[Note: Customer Expectations: Speed of Payment, ease of use, variety of choice]



These cards may be used at ATMs, POS machines, shops, online transactions, wallets, for e-commerce portals and micro ATMs.

Debit Cards

In India, 32 lakh cards were illegal. Indian money was drawn in China and America. This news came in Indian newspapers on 21st October, 2016. Only pin change and new cards are safe. The main banks who involved in such things were SBI, HDFC, ICICI, UCO, Yes and Axix New cards. SBI and other new banks have come forward to provide new cards. The remedy is get a new pin number on your existing card or obtain new debit cards. Even on Internet Banking also, pin change. Canara Bank asked the account holders to change the pin immediately, otherwise it will be blocked.

The number of cards affected: 26 lakh Visa Master and 6 lakh RuPay.

10.4 RECEIPT OF PAYMENTS

These are from three sources:

- Retail customers
- Wholesale customers and
- Various miscellaneous payments

Before the advent of online electronic payment methodologies, all the payments were received generally in cash or cheques and periodically the organisation would make a physical deposit at a bank. New electronic methods have made it simpler and faster through the use of wire transfer making the funds transfer possible without a physical deposit.

10.5 THE TRADITIONAL PAYMENT SYSTEM

This system uses currency, coins and cheques. Currency and coins can be used to mediate transactions by physical exchange. Even though cash is legal tender for the payment of debts, it leaves no audit trails. Many people ask for written receipts to show proof of payment when they pay a debt or purchase goods with cash. As an alternative to cash, many people write cheques that tell their depository institution to transfer funds to the person/enterprise named on the cheque.

The advantages and disadvantages are there in the cheque clearing process, cheque float and cheque information.

The other systems which are recently in use are:

- Value exchange systems.*
- Shared terminal systems.*
- Credit card systems.*
- National networks.*
- Electronic funds transfer (EFT) systems.*
- Non-bank networks.*
- Automatic deposit of payrolls and social security payments.*
- Bill payments by telephone.*
- GIRO transfers.*
- Point-of-sale payments systems.*

Some of these are explained below:

In addition to this, home banking has developed in many countries because of the reduced cost of computer and communication technology. In home banking, computers use connections over telephone lines or cable TV networks to access a depository institution's account records. Most allow people to check their account balances, pay their bills (to cooperating merchants) and transfer funds among accounts. Many also allow people to keep home budget, accounting and tax records in an organised fashion. Home banking devices frequently are used to access computerised information service systems. Such networks can provide stock and commodity market quotations, electronic shopping, current news, weather reports, airline travel information, computer games and a host of other services. People who use a home banking network can select an item to buy from an electronic catalogue, order the goods, transfer funds from a savings to a checking account to replenish its balance. They may also be able to record the item in the proper home budget category and note whether it is a deductible item for tax planning.

Value Exchange Systems

These can work much like cheque clearing systems, although they transfer assets other than claims on demand deposits. The first widespread development of value exchange involved the use of credit cards. In such system, credit card holders' IOUs were exchanged for cash balances. Subsequently, these have been used to transfer debits to both demand deposits and other accounts. Accounts used for value exchange, in addition to demand deposits and credit card lines, include balances in individual's margin accounts held with stock brokers, balances in accounts held with money market (or even long-term), mutual funds and accounts held with other financial institutions.

The key element necessary for a value exchange is that of the value of the item exchanged can be readily determined and agreed upon. IOUs of credit card customers and margin account customers have a fixed monetary value, as do shares in money market mutual funds (MMMF). Shares in other funds can also be exchanged if fractional share withdrawals are allowed when the net asset value per share is not equal to even a rupee.

Credit Card Systems

These initially used sales receipts are signed at the point of purchase. After making a phone call or verifying an individual's identity or signature, a merchant would accept the credit card receipt in exchange for goods and services. The receipts would then be forwarded to the institution that issued the card (usually a specialised credit card receipt clearing system). The card issuing institution would then add the amount of sale to the individual's balance and send the credit card user a bill. Under country club billing systems, the credit card issuer would send a copy of the receipt along with the bill so that the card holder could verify the debt.

The physical transfer of credit card receipts was very expensive. Thus, credit card systems were among the first to develop computerised transfers of information and computerised billing systems. VISA (initially Bank Americard) and Interbank Card Association (initially Master Charge) developed computerised information network to transfer credit card billing information around the country (in USA).

Computerised billing allowed credit card receipts to be 'truncated'. A merchant who accepted a VISA or Master Card receipt in return for a sale could take the receipt to a cooperating bank in the appropriate card system and receive immediate credit (minus a merchant discount). The receiving bank would pass the receipt on to a clearing bank in the credit card system. That bank would then retain the credit card receipt and transmit all pertinent information (consumer's credit card number, amount of the charge, the date, the place and the merchant) through the credit card data processing network. The processing bank, in turn, would receive credit through the credit card clearing network for the funds it disbursed, plus a fee (possibly paid in the form of a smaller discount) for its service. Since the credit card was retained by the processing bank, its physical movement was said to be truncated. However, the pertinent information on the receipt was passed on electronically.

When the electronic information reached the bank that issued the credit card, it would debit the customer's account (increase the balance due owed by the consumer) and, at the end of the billing cycle, bill the customer for the full balance due, plus any applicable financier's charges. The bill would indicate the amount of each charge and its date and place. This process is illustrated in Fig. 10.6. The value that was exchanged through the clearing system, then, was the value inherent in the consumer's willingness to pay his debt. The card systems also developed information transfer, clearing and funds transfer technology necessary for Electronic Funds Transfer systems.

The clearing information and facts through value exchange system are given below.

10.6 ELECTRONIC FUNDS TRANSFER (EFT) SYSTEMS

There are a wide variety of uses of electronic funds transfer (EFT) technologies. Other uses often require automated clearing houses (ACHs). One of the first uses was for the transfer of computerised credit card clearing information. The most sophisticated EFT operations are geared to ACHs. ACHs can provide either:

Credit transfers – the initiating institution sends funds through the system to be deposited in the recipient's account.

Debit transfers – the initiating institution withdraws funds from the depositor's account.

Examples of credit transfers are automatic deposit of payrolls, bill payments by telephone and GIRO transfers (a type of payment used in Europe).

Examples of debit transfers are pre-authorised bill payments, transfers initiated by point of sale (POS) transactions and cash withdrawal from remote shared terminals. Bill-cheque payment mechanisms can be adapted to either credit or debit transfers.

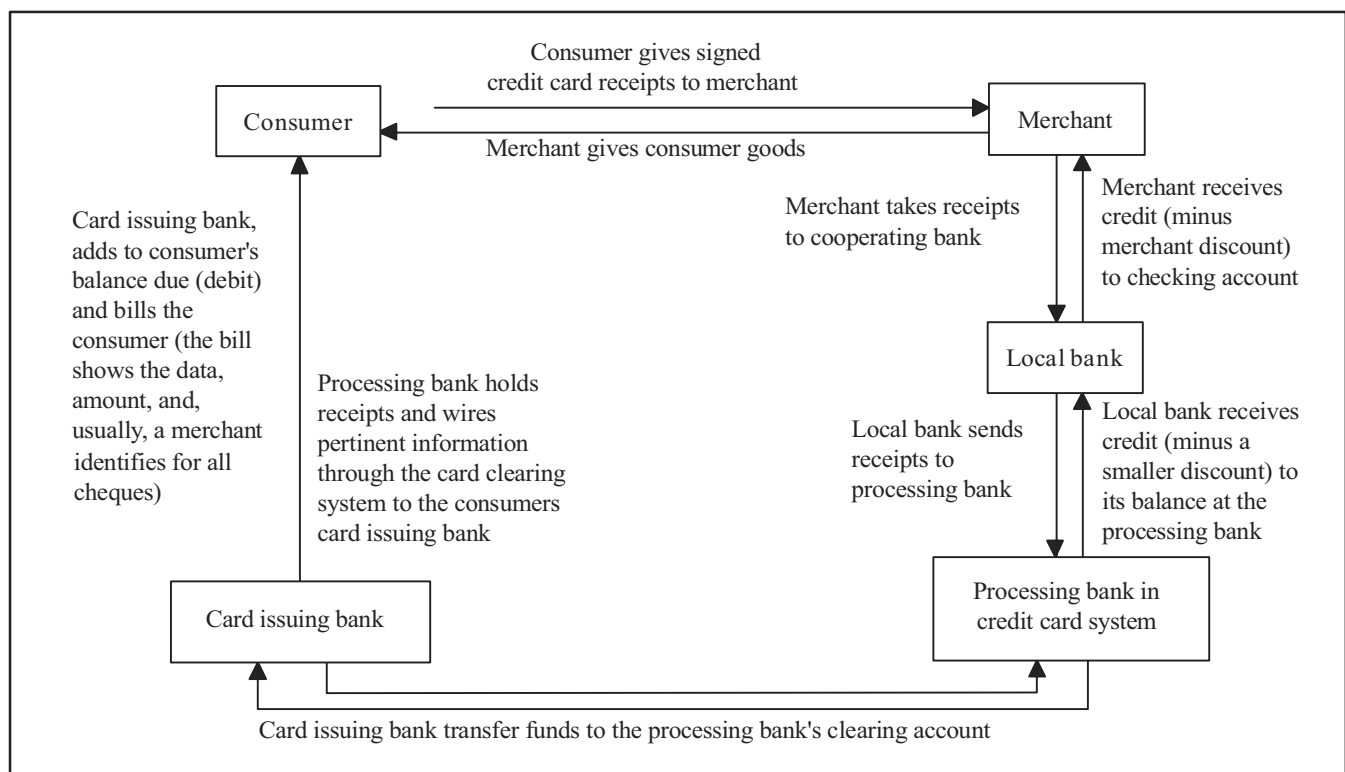


Fig. 10.6 – Clearing information and facts through a value exchange system

Paperless Bills

You may be able to pay all bills at a single website. Indian consumers also nowadays are flooded with bills almost everyday; bills for utilities, bills for credit cards, bills for medical expenses, bills for electricity/water, bills for house taxes, bills for income tax, bills for hotels, choultries, bills from post offices, bills for housing rent, bills for rentals of various items, etc. The paper trail of bills, cheques and receipts is confusing at best, but the Internet is offering a solution, electronic bill presentment and payment.

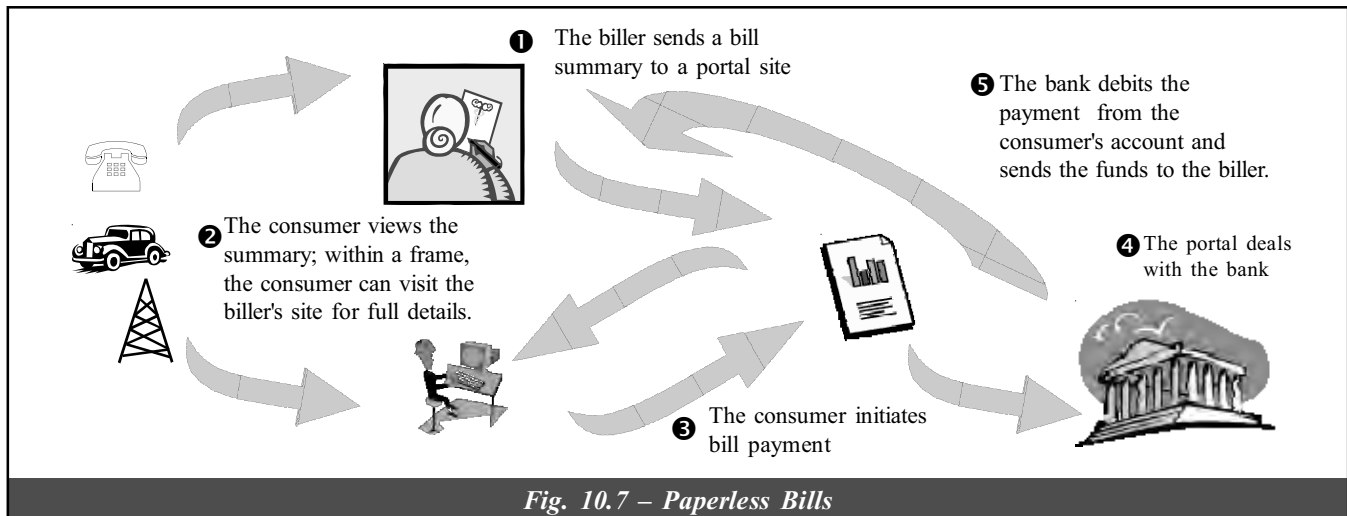


Fig. 10.7 – Paperless Bills

Credit and Debit Transfers through an ACH

Electronic credit transfer funds and information simultaneously from the payer to the payee. Debit transfer funds from the payer's bank to the payee's bank, but only after the payer gives the payee valid authorisation to do so.

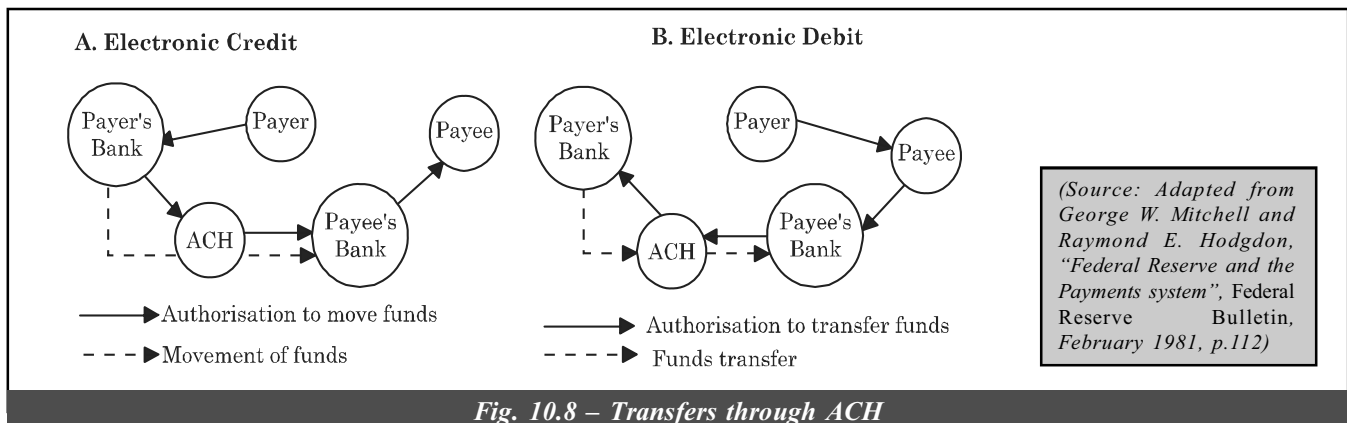


Fig. 10.8 – Transfers through ACH

Bank Account reconciliation systems are changing as EFTs extend to bank customers. There will be a need continuously for such systems due to ensuring of data integrity.

Automatic Deposits of Payrolls and Social Security Payments

Regular payments to the same individuals can be made efficiently through automatic clearing houses. As illustrated in electronic credit, the bank of the paying organisation gives the ACH a computer tape that provides information on the banks and account numbers of the firm or agency's employee (or government pension and social security recipients) who are to be paid. The clearing house then credits the accounts of the receiving banks (or other financial institutions) with the total due and provides data indicating which depositor's accounts should be credited. The bank that submits the tape provides funds to the clearing house if its payments exceeds its receipts at the clearing house from other sources.

For ACH operations to succeed, all organisations that submit tapes must use the same format and method of identifying the account numbers and amounts of funds transferred. The advantages of credit transfers are that they are fast and allow both payments and related accounts information to make only one trip through the payments system.

Bill Payment by Telephones

This can be made efficiently through ACHs, using telephones or cable television lines. To make payments, the depositor dials the bank, enters his account number, the amount he wishes to pay, and the bank and account numbers of the recipient of the payment. If the depositor provides data electronically (touch-tone phone), the payment information can be verified and entered directly on a magnetic tape for submission to ACH, which will make the appropriate credit transfers.

GIRO Funds Transfer Systems

These are widely used in Europe. It occurs when an individual instructs a financial institution to make a payment to another individual (or company) who banks with another institution. The originating institution then transfers both funds and payment information telling which account should be credited to the receiving institution. Thus only one transfer is required for both funds and transmitted information. EFT systems can make similar transfers if appropriate payment information is entered on magnetic tape and sent to an ACH.

Point of Sale Payments System

These systems are more complicated than payroll, pre-authorised transfer, or bill cheque payment systems. They require a two-stage transfer of information and funds. One transfer requires that information on the status of the customer's account be transferred to the point of sale so that the expenditure can be authorised.

The second transfer requires that funds be transferred from that account (and the institution that holds it) to the retailer's account (and the institution that holds it). Because of the complexity of dual information and funds transfer function, this system has been slower to develop than other EFT functions. However, because of greater convenience to the consumers, numerous financial institutions prefer this system.

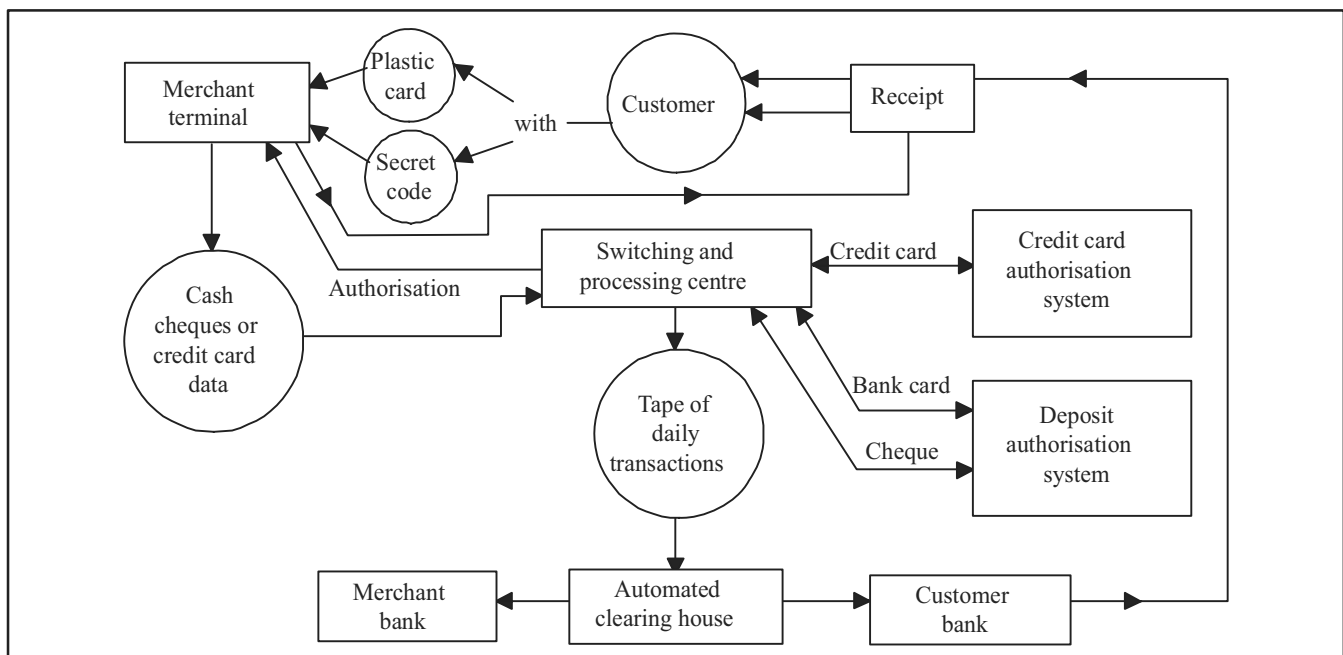


Fig. 10.9 – Point of sale EFT transfers

Point of sale transfers are more complicated than pre-authorised transfers. First, a customer must prove that he is authorised to transfer funds from the named account. Usually, this is provided with a personal identification number or a secret code at the point of sale. Second, the merchant must verify that sufficient funds are available to be transferred. This is done by contacting a credit card or debit card authorisation centre that has a record of the customer's account. The actual funds and goods are transferred only after the merchant receives the authorisation to do so.

10.7 MODERN PAYMENT SYSTEM

These are done through:

- PC Banking
- Credit Cards
- Electronic Cheques (i-cheques)
- Smart Cards
- E-Cash
- Micropayments

(i) PC Banking

This has been developed after the introduction of PCs and modems. PC banking involves externally accessible bank systems.

Earlier, payments were made to companies by retail customers through direct debit (automatic withdrawals from the customer's bank account) or ATMs (automatic teller machines). Today, there are numerous payment options.

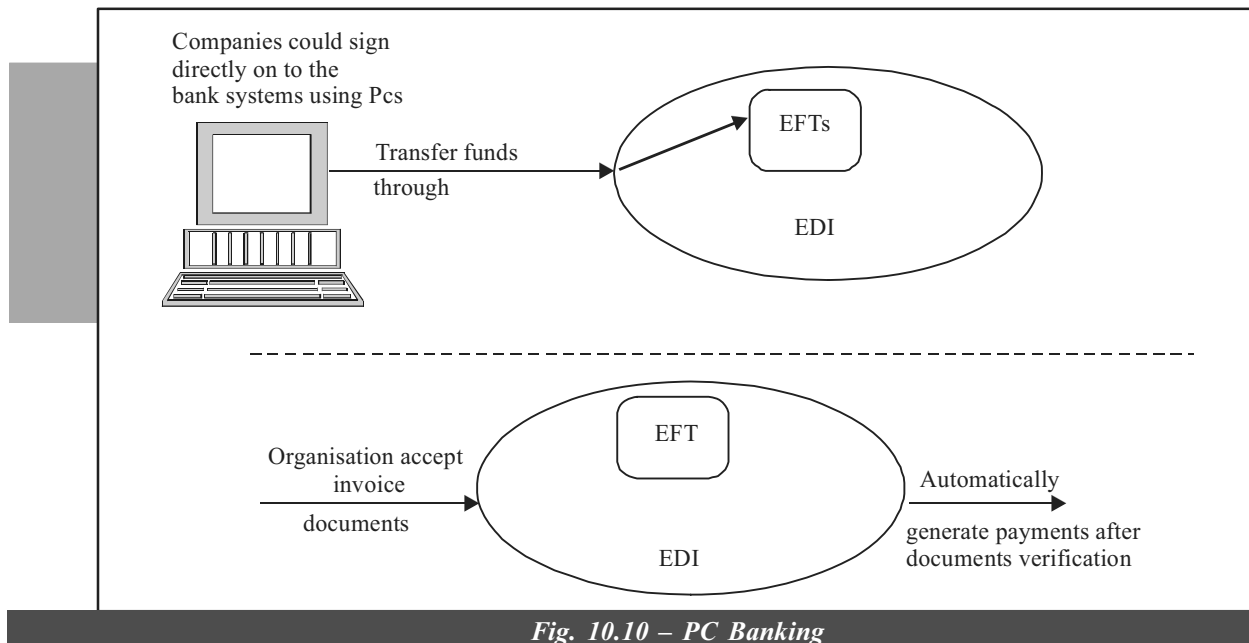


Fig. 10.10 – PC Banking

(ii) Credit Cards

History starts in using credit cards in 1958, when the Bank of America issued certain cards in California to select group of users to pay for products by participating merchants, without cash. The advantage of using such cards is that users no longer have to carry cash to pay for a product. Earlier, a credit card was provided imprinting a number. This was sufficient to complete the transaction. Electronic processing was found necessary for online authorisation and depositing functions. In the late 90s, the Internet provided an unlimited opportunity if businesses could get paid for products or services sold electronically. The inevitable marriage of the two technologies was a match made in virtual heaven. Credit card numbers, user names and expiration dates were provided over the Internet in insecure forms. This risky process was updated to include measures that ensures that the credit card of yours is not visible to other users (encryption) and a way to ensure that the party you are sending your information is the only company receiving it (authentication).

To accept these cards as a payment option, one has to sign on as a merchant with your bank or credit card organisation. A percentage of fee may have to be paid depending on the size and the products sold. There are a number of vendors on the Internet who will process the credit information for you once it has been entered into their website.

The credit card processing is simple. After receiving from a customer details like his credit card number, name, expiration date, two more things should occur to complete the transaction.

- The credit card must be authenticated with the credit card company so that you can receive an authorisation number. This provides guarantee that the funds will be made available to you when you complete your deposits by ensuring the user of the card is not over his credit limit and that the card is not stolen.
- The electronic credit card information must be turned into an actual deposit at your financial institution.

Cyber Cash, one of the companies operating in the above field, provide Internet payment processing software, accepts and processes credit card information to receive online authorisations. They provide software for linking to the financial institution to credit electronic credit card deposits.

(iii) Electronic Cheques/Internet Cheques (i-cheques)

Electronic cheques also known as i-cheques are used to make electronic payment between two parties through an intermediary and not very much different from the current cheque processing system. Here, the electronic cheques are generated and exchanged online. The intermediary will debit the customer account and credit the merchant account.

(iv) Micropayments (Electronic Wallet or Microtransactions)

A micropayment is a very small fee paid when a visitor is sent from another website to your site. The payment is usually tracked by a third party and when a specific amount has been accumulated, say rupees thousand, the fee is paid to the website that sent the traffic. This payment mechanism helps a way for you to encourage other companies to add to your links into their website. If any of the customers of other websites enter their site and click the link, the click through to your website will be recorded. In view of the services provided, when enough visitors click through to your website have been accumulated, a payment will be made to the company that sent the traffic. It is cash payment which is made as a result of using a service on the Internet. These are also made between two or more Internet companies to help generate traffic through links. A micropayment work based on the registration made within an organisation that verifies and reports on traffic levels between sites. Next, a very small payment per link or per click is calculated and when the total owing reaches a specified threshold, the payment is released from the customer's actual or virtual bank account to the vendor's account.

The problems generally occur with payments based on click-throughs is that some web masters have found ways to defraud the system. The main reason for using a third party to track pay-per-click micropayments is to reduce the potential of fraud. These organisations will manage the security aspects of these programs to ensure that the traffic generated is not repeat traffic or that it has not been artificially created by company.

Carnegie Mellon University's Net Bill supports micropayments. Cyber Cash is also spearheading micro transactions. Web purchases with \$10 are not practical using credit cards because of processing cost. Cyber coin system uses encrypted digital signals in the place of real coins. Microsystem can be divided into:

- Debit/credit (pay earlier/pay later), e.g., Net Bill system
- Digital cash (pay now)

A micropayment is needed for pricing microproducts and microbundles. Fee for this are very less and will become largely unimportant. If transaction costs can be made low enough, why should digital product sellers be limited to accept credit card payment and other large-scale payment methods. Micropayments are as essential to electronic commerce as product customisation. Micropayment and unbundling will be a natural response to a growing abundance of customised microproducts. The New York Times or CNN may not be the sole gatekeepers of news, ideas, information and knowledge. The Internet is a natural dispersed information channel for small, occasional sales. Its unique interactive capability also makes everyone a potential guest columnist on the net. These microproducts will be paid for not by subscription but by micropayments. Short-term sales (one page of information) will be based on micropayments. Short-term sales may produce higher quality than subscription and bundling.

(v) Smart Cards

A smart card can be considered as a portable device that contains some non-volatile memory and a microprocessor. This card contains some kind of an encrypted key that is compared to a secret key contained on the user's processor. Some smart cards have provision to allow users to enter a personal identification number (PIN) code. These are being relatively common with billions of cards expected to use worldwide. These will emerge as the ultimate interface device for the mobile digital economy. These will hold your cash, ID information, house and office keys, subway tokens, all types of preference files (house temperature setting, driver seat setting, etc.) and other information. You will exchange these information and digital products with other people, transact business, present to police officers, check into a hotel or a sports arena and other areas unimaginable.

The smart card technology is widely used in countries such as France, Germany, Japan and Singapore to pay for public phone calls, transportation and shopper loyalty programs. The idea has taken more time to catch on in the US, since highly reliable and fairly inexpensive telecommunication system has favoured the use of credit and debit cards.

Smart cards are of two types:

- Relationship based smart credit cards and
- Electronic purses (replace money; known as debit cards and electronic money)

Traditional credit cards are fast evolving into smart cards as consumers demand payment and financial service products, that are user-friendly, convenient and reliable.

Relationship based smart card is an enhancement of existing card services and/or the addition of new services that a financial institution delivers to its customers via a chip-based card or other device (includes access to multiple financial accounts, value-added marketing programs or other information card holders may want to store on their card). These are credit-based and settlement occurs at the end of the billing cycle inspite of its increasing flexibility as the advantage.

Electronic purses are wallet-sized smart cards embedded with programmable microchips that store sums of money for people to use method of cash for everything from buying food, photocopies, subway fares, etc. These are financial instruments to replace cash. To meet this need, banks, credit card companies and even government institutions are racing to introduce these.

A smart card, unlike a credit or debit card, is a card that is embedded with either a microprocessor with a memory chip or only a memory chip with non-programmable logic. While the microprocessor can add, delete, and otherwise manipulate information on the card, a memory chip card can undertake only predefined operations.

The origin of smart cards can be traced to a Frenchman named Ronald Moreno, way back in 1970.

As on date, there are three categories of smart cards:

- A memory card that stores data. It will need an outside processor to access and work with this data.
- An intelligent memory card, which contains besides data, some built-in logic, usually used to control access to the memory of the card. These cards have certain data processing capabilities.
- A processor card with its own microprocessor and an operating system. These cards can process and store data on their own.

Two other types of cards are:

- Contact smart cards are inserted into a reader or validator for the data to be read. When the card is inserted into the reader, it transfers data to and from the chip via electrical connectors.
- Contactless cards are passed near an antenna to connect via a radio signal. They have both a microchip and an antenna embedded. This allows the smart card to communicate without physical contact.

Further, research is also going on into applying biometrics in the design of smart cards. It is possible to store an image of an individual's body part say his face, fingerprint or even an eyeball, in memory and invoking it, and comparing it with the corresponding image of the user. A smart card of the future can use this technology to detect impostors and frauds. Kodak and Finansforum (a Swedish firm) are the two global companies doing advance research in this area. Some of the leading manufacturers of smart cards are Hitachi America, Motorola, Philips semiconductor division, Racom Systems Inc., SGS Thomson and Siemens.

10.8 ELECTRONIC CASH

The predominance of cash indicates an opportunity for innovative business practice that revamps the purchasing process where consumers are heavy users of cash.

Properties of E-Cash

E-cash have the following four properties:

- Monetary value
- Storability and retrievability
- Interoperability and
- Security

E-cash is based on cryptographic systems called digital signatures. A pair of numeric keys (very large integers or numbers) work in tandem – one for locking (encoding) and other for unlocking (decoding). Messages encoded with one numeric key can only be decoded with the other numeric key, not otherwise, the encoding key is kept private and the decoding key is made public.

By supplying all the customers (buyers and sellers) with its public key, a bank enables customers to decide any message (or currency) encoded with the bank's private key. The customer would be happy and confident if decoding by him yields a recognisable message. He understands that only the bank could have encoded it. These digital signatures act as security (as mathematics involved) and have proved successful over the past two decades to be more resultant to forgery than handwritten signatures. The currency server provides e-cash to the buyers and the purchase from an online currency server (or bank) could be done in two steps:

- Establishment of an account
- Maintaining enough money in the account to back the purchase

Customers may even buy e-cash by paying paper currency, to maintain anonymity or because they don't have a bank account. Consumers use the e-cash software in the computer to generate a random number, which serves as the "note". The bank uses its private key to digitally sign for the amount requested and transmits the note back to the customer. Thus, a bank note, with a serial number and a rupee amount indicated and with digital sign is obtained. The bank with digital signature is committing itself to back that note with its face value in real rupees. The method of note generation is highly secure as neither payer nor payee can counterfeit the bank's digital signature (similar to the watermark in paper notes). One drawback of e-cash is its inability to be easily divided into smaller amounts. However, it fulfils two main functions: (i) as a medium of exchange and (ii) as a store of value.

10.9 THE STEPS FOR ELECTRONIC PAYMENT

First, it is important to put up your website on a server. In e-commerce terminology, this is called as a Commerce Server. This is a server that manages and maintains all transactional and back-end data for commerce website. In India, hardly anyone hosts their own site. This task is often not outsourced. There are a number of web hosting providers available. The hosting service should be capable of handling much greater traffic than that expected initially. Also, check if the servers of the hosting service provider are not non-functional often. This seems to be a problem with many service providers. Most of those providing hosting in India also ultimately rely on service providers in the US, etc. So, minimise the intermediaries and don't end up signing a sub-sub-agent.

Nowadays, businesses need to respond to large volumes of e-mail daily. This requires a robust management system that sorts, sifts and responds quickly and correctly. This can be achieved by outsourcing this function. Many specialised softwares are available to handle e-mails and could get the job done in most cases. Hence, opt for outsourcing this function only if you have large volumes of e-mails to deal with. In-house management is always better as privacy can be taken care of and you stay closer to the customer.

When you are conducting business transactions electronically, online payment would require linking up to processing banks. A large number of such businesses rely on a payment outsourcer. Sign up a payment outsourcer who accepts all major credit cards and employs reliable fraud protection and encryption. This is a crucial function in e-commerce and has to be handled with due care.

Products ordered from a website are tracked through the payment process to the warehouse and shipped to the customer. The delivery mechanism has to be designed considering the area you intend to cater to and the time duration between the order being placed and actual delivery. Your business could be much more hassle-free if this function is outsourced. Finally, the most vital aspect of e-commerce is that of security. Security is an issue that cannot be compromised by any e-business. Regular checks and audits are essential to ensure the site security. This requires special tools and skills and could be outsourced; though there are not many who tackle only security-related issues in India. One lapse in security can ruin your business forever. So, put your site through rigorous security testing before opening shop. Once you have put up a site of your own, instead of just providing a phone number or e-mail for the potential customer to contact, it would be much better if the customer could place an order then and there on your website itself. Studies have shown that very few potential customers actually follow through and buy goods from a site, which does not provide a speedy and easy way of buying online. To allow your customer the additional option of ordering directly online with a credit card while he or she is viewing your site, you need to follow certain procedures to see your business through. First, get a merchant account. Merchant account is an industry term meaning a banking relationship, which allows a business to accept credit cards.

In other words, an Internet Merchant Account is a special account that permits the acceptance of credit cards online. Transactions are processed online in real time. To make e-commerce hassle-free, you can outsource whatever portion you need if you don't have the expertise in-house. Outsourcing is the utilisation of a third party service company to provide you with those services that complete the total functionality of the business process. Then, as mentioned earlier, security is always the prime concern when it comes to any transactions on the Internet. It is important to follow established conventions in Internet security, which means utilising a secure server for credit card transactions.

A secure server is one running software capable of establishing a "secure" connection with your customer's Internet browser using SSL (secure socket layers) technology which encrypts all transmitted information. *Secure Socket Layer is an encryption technology on the server that scrabbles important data such as credit card numbers and order information when it is being stored or passed from one computer to another.*

Most net buyers insist on this connection before entering credit card information. You would not need a secure server if you were outsourcing your order processing. Equally important is to present all your products, prices and options (sizes,

colours, etc.). Before you can request credit card information, you ought to automatically subtotal the order, add shipping and handling and calculate taxes to be levied if any. After the customer furnishes personal details and credit card information along with his order, there should be effective online clearing. This means authorisation of the customer's credit card while he is online.

This feature is applicable if you are providing purchased software for downloading. Designing online clearing capability is extremely complex and thus this option is best undertaken through an outsourcing solution. Finally, a proper system has to be put in place to track the orders that are received and keep a record of the details of each transaction.

The online payment follows a simple procedure and is easy and uncomplicated enough for any new entrepreneur to understand. It begins once the customer makes the purchase decision and complies with the prescribed mode of purchase on your website. Once an item is purchased using a credit card, you or the automated process on your website requests an authorisation from the customer's credit card account to ensure that the card is valid and the customer has the balance available for the purchase. The customer's bank then sends an authorisation number back to you to your website and the order is "captured". After confirmation of the order, the merchandise order is sent to the customer. You then inform the bank of your account information so that the bank can deposit the funds into your account. Now, the order is sealed and transaction is complete.

Payment Alternatives

Another kind of transaction that can be carried out without the direct use of credit card or disclosing the credit card number to the merchant is by Electronic Cash Transactions. Electronic money is an arrangement whereby the customer pays for the merchandise using electronic money. Examples of this are Cyberbucks, CyberCash, etc. As consumers become more comfortable providing credit card information over the Net, these methods should be less utilised in future.

The working of CyberCash is a lot more different and more complicated than that of payment by credit cards. The CyberCash system extends the well-trusted automated POS paradigm to online transactions.

The main components of this system are:

- (i) The CyberCash Wallet distributed free to consumers
- (ii) The Secure Merchant Payment System (SMPS) for Cyber-merchants
- (iii) CyberCash Gateway Servers linking to existing financial networks
- (iv) CyberCash Wallet – Universal payment in Cyberspace.

The CyberCash Wallet is software that resides on the consumer's personal computer to enable secure Internet payment. *It is universally available – the software can be downloaded free of charge at any time from the CyberCash website, or from merchant sites at the time of a purchase. Basically, electronic wallets store your credit card numbers on your hard drive in an encrypted form. You then make purchases at websites that support that particular type of electronic wallet. By clicking on a Pay Button, customers initiate a credit card payment via a secure transaction enabled by the electronic wallet company's server.*

The Secure Merchant Payment System (SMPS) is a software package distributed by CyberCash to merchants. *The SMPS resides on the merchant's web server, interfacing with a customer's CyberCash wallet on the front end.*

The actual working of CyberCash is a lot more routed than that of credit cards. It can be explained in the following example.

A customer visits your website using standard web browser software and fills his shopping cart. You then exchange details on addresses, delivery and final price – all online. Your web server then displays the CyberCash PAY button to the customer, initiating a transaction. The customer clicks on the PAY button, prompting your web server to send an electronic invoice. The customer's CyberCash wallet automatically opens, allowing him to select a payment instrument. An encrypted charge payment message is then sent to your web server. When your SMPS software receives the payment message, it adds your merchant identification information. The payment request is then forwarded to the CyberCash Gateway Server through the gateway to the bank. The CyberCash Gateway Server decrypts the message and authenticates your customer's CyberCash wallet and you as a valid merchant. A message is then sent (using secure, private financial networks) to your bank or its authorised processing centre requesting approval.

Your bank responds accordingly. Once the request is processed, a positive or negative response is sent back to your server and on to your customer. If positive, your server receives a digital receipt. If negative, your customer can select an alternative payment option. So, you don't necessarily lose the sale.

For an interactive illustration of the working of CyberCash, check out – www.cybercash.com/cybercash/shockwave/anatomy-shock.html which provides for a shockwave presentation of the working.

10.10 PAYMENT SECURITY

The payment information can be stolen or altered in a number of ways by the criminals. The intent is to capture personal financial information like credit card numbers, the number on the card, the name on the card and the expiration data. Once this is found out, the information can then be used to purchase other goods online for delivery to a temporary address. By the time the fraud is detected, the criminals would have disappeared from the spot. The techniques involved in this are:

- Spoofing
- Sniffing
- Content alteration and
- Denial of service

(i) Spoofing

Hackers can spoof or configure a system to masquerade as another system, thus gaining unauthorised access to resources or information on systems that 'trust' the system being mimicked. *A hacker may assume someone else's identity, whether it be a login ID, an IP address, a server or an e-commerce merchant or he may perform the act of mimicking a legitimate website including layout, colours and function to obtain credit card information or to steal business. The technique involves:*

- Registering a domain name that is very similar to a legitimate sales site, perhaps different by a single letter such as Amazon.com (in USA) or hnt.com (in India) for Amazon and hmt respectively.

The illegal website then copies the text and graphics of the so-called legitimate site and certain functionalities of the legitimate site are built to mimic the feel of the links connected in the site. It later on offers a popular product at a ridiculous price to entice people to send their credit information. Though it is generously known that not all sites that mimic using the misspelling technique are bad, there are a few who creatively divert businesses to their legitimate site. Others are simply capitalising on popular phrases. There are legitimate spoof sites who legitimately own the domain name of a popular website and place a notice indicating that is not that popular site which is operating.

(ii) Sniffing

If a hacker has gained access to a host, the hacker may set up sniffing programs to observe traffic and storing information (IDs/Passwords) that can be used to compromise other systems. It is unauthorised network monitoring. Sniffer attacks begin when a computer is compromised and the cracker installs a packet sniffing program that monitors the network to which the machine is attached. The sniffer program watches for certain kinds of network traffic; typically for the first part of any Telnet, FTP or login sessions – sessions that legitimate users' initiative to gain access to another system. In the first part of the session, login ID, password and user name of the person logging into another machine, all the necessary information a sniffer needs to log into other machines are done.

But as the days progress, the sniffer gathers all information about local users logging into remote machines. One insecure system on a network can expose to intrusion not only to other local machines but also any remote systems to which the users connect.

Sniffing describes the act of reading unprotected packet information as it travels over a network. *The act of sniffing packets is relatively simple with the use of various network sniffing network programs that were originally designed to assist in network error detection. Sophisticated hackers lurk or watch the message to and from certain IP addresses, and if they find something, they sell the information to criminals or competitors. On each machine, connected to the network software called a protocol operates as a network language (of sorts). This, combined with a piece of hardware called a network adapter (connector to the network) and a little configuration, will result in messages being sent to or received from another computer. "To" and "From" IP addresses are attached to each message sent on the Net. All routers look at or will analyse and hand off to a web server those messages that pass by until they locate their own IP address. The route will analyse and hand off to a web server those messages that will be allowed through the network firewall.*

(iii) Content alteration

A method involved in diverting payments would be to intercept a bank account number and change it to another. This is called content alteration. *Sniffers, who are involved in this, capture the packets, after the content and then send the new packet to its intended recipient. Shipping address for an order is also an alteration made in the piece of information.*

(iv) Denial of service

Servers can be attacked with threats such as denial of service, where a user can render the system unusable for legitimate users by “hogging” a resource or by damaging or destroying resources, so that they cannot be used. The two most common forms of denial-of-service attack are:

- Service overloading and
- Message flooding

Servers are vulnerable to service overloading. One can easily overload a www server by writing a small loop that sends requests continually for a particular file, e.g., a homepage. The server tries to respond to the request in good faith as it assumes that all requests are legitimate. Denial-of-service attacks may be caused intentionally or unintentionally by ‘runaway’ software programs such as those caught in an infinite loop.

Message flooding occurs when someone sends a very large file to a message box every few minutes. *This box grows in size quickly and begins to occupy all the space in disk and increase the number of receiving processes on the recipient’s machine, typing it up even more and thus causing a disc crash. To avoid this, it is necessary to provide separate areas for different programs and make provisions for graceful failure. This is the best solution.*

This illegal technique involves conducting business with an organisation electronically and then denying that the transaction occurred, or denying it was ever initiated. An Internet user may order product on a credit line and have it drop shipped to another location. When the bill is received, the user denies that he never ordered the product.

(v) E-cash

This can be thought of as the minting of electronic money or tokens. In electronic cash schemes, buyers and sellers trade electronic value tokens which are issued or backed by some third party, be it an established bank or a new Internet-based institution. The effects of a system failure in an electronic cash scheme are much harder to anticipate. System failure could also occur through many means, not the least of which is insufficient funds (or paper money) to back-up the electronic money.

For example, DigiCash offers a token system, where the actual money resides on the buyer’s hard disk. Most e-cash systems have an audit trail in their systems for reasons of security to allow a transaction or e-cash value of money to be back-tracked to the source, irrespective of the number of times the unit of currency is traded between e-cash users. The company addresses the issue of privacy.

A DigiCash user requests a supply of e-cash tokens from the DigiCash server; those tokens are encrypted using a proprietary encryption system and e-mailed to the holder. If the holder buys something thereafter from a company or another e-cash account holder, the buyer mails the tokens to the seller, who then banks them with DigiCash. Here, the recipient knows where the tokens have come from, and the sender knows where the e-cash was sent. DigiCash only knows that the e-cash has been banked by the person or organisation. DigiCash knows only where the e-cash has come from directly, but it does not know where the e-cash originated from. This system is called one-way privacy, because, none, not even a legal authority, can back-track the chain to find out who has the e-cash and where it has come from. DigiCash’s e-cash system is now provided by Mark Twain Bank in the USA. EUnet, the largest European ISP, also provides e-cash in cooperation with Merita Bank, Finland’s largest commercial bank.

Electronic cash is a new concept in online payment systems because it combines computerised convenience with security and privacy, as a result of which it improves on paper cash. It opens up a host of new markets and applications in view of its versatility. It is an attractive alternative for payment over the Internet by replacing cash as the principle payment vehicle in consumer-oriented electronic payments.

Cash remains the dominant form of payment for three reasons:

- Lack of trust in banking system
- Inefficient clearing and settlement of non-cash transactions and
- Negative real interest rates paid on bank deposits.

10.11 INFORMATION SOURCES FOR ELECTRONIC COMMERCE AND PAYMENTS ON THE WEB

Following are the different sites, addresses and descriptions as taken from PC Today magazine (www.pc-today.com).

Site	Address	Description
Center for e-commerce	http://www.iti.org.80/cec/	Part of the Industrial Technology E-commerce Institute (ITI) at Ann Arbor, Mich, supports enterprise integration by helping industry adopt EDI.
Commerce Net	http://www.commerce.net	An industry consortium addressing the needs of business for EDI and electronic commerce over the Internet.
CyberCash	http://www.cybercash.com	Developer of secure Internet payment software and services.
DigiCash	http://www.digicash.com	Developer of digital cash software and services.
First virtual	http://www.fv.com	Provider of Internet payment services.
The Internet commerce connection	http://www.loshin.com	Includes pointers to electronic commerce resources and experts from books about doing business online.
Mastercard International	http://www.mastercard.com	Provides information about the SET standard as well as the credit card associations activities'
VISA International	http://www.visa.com	

Table 10.1 – Information Sources For Electronic Commerce

(Source: Adapted from Peter Loshin, "The Electronic Marketplace", *PC-Today*, July 1996, pp. 85, 88).

10.12 INTERNET PROTOCOL SECURITY (IPSEC)

Internet protocol security is a suite of protocols used for securing private communications over the Internet. The proposed suite of IPsec protocols would create a standard platform for securing IP connections on private networks.

The Internet is inherently insecure and hence business still needs to preserve the privacy of data when it travels over the network. In view of this, the Internet Engineering Task Force is an international group of network designers, operators, vendors and researchers concerned with the evolution of Internet architecture – has developed a set of protocols called Internet Protocol Security (IPsec). IPsec creates a standard platform to develop secure network and electronic tunnels between two machines. Secure tunnelling via IPsec creates circuit like connections via network through which data packets can move. It creates these tunnels between remote users and within a local network. It also encapsulates each data packet in a new packet that contains the information necessary to set up, maintain and tear down the tunnel when it is no longer needed.

Encryption is used to ensure the confidentiality, integrity and authenticity of the two end points in the private network. The Internet Key Exchange (IKE), an application layer protocol, authenticates each peer in an IPsec transaction. IKE negotiates security policy, determining which algorithm may be used to set up the tunnel. Further, it handles the exchange of session keys used for that one transaction. Networks that use IPsec to secure data traffic can automatically authenticate devices by using digital certificates which verify the identities of the two users who are sending information back and forth. IPsec can be an ideal way to secure data in large networks that require secure connections among many devices. IPsec allows interoperability among different network devices, PCs and other computing systems. The Protocol Suite is available as a software only to upgrade the network infrastructure, by allowing security to be implemented without costly changes to each computer. IPsec is also useful when setting up remote users. These workers can use an IPsec client on their PCs in combination with other secure tunnelling protocols such as layer 2 tunnelling protocol, to connect back to the network. This reduces the cost of remote access and offers greater security than is possible over dial-up lines.

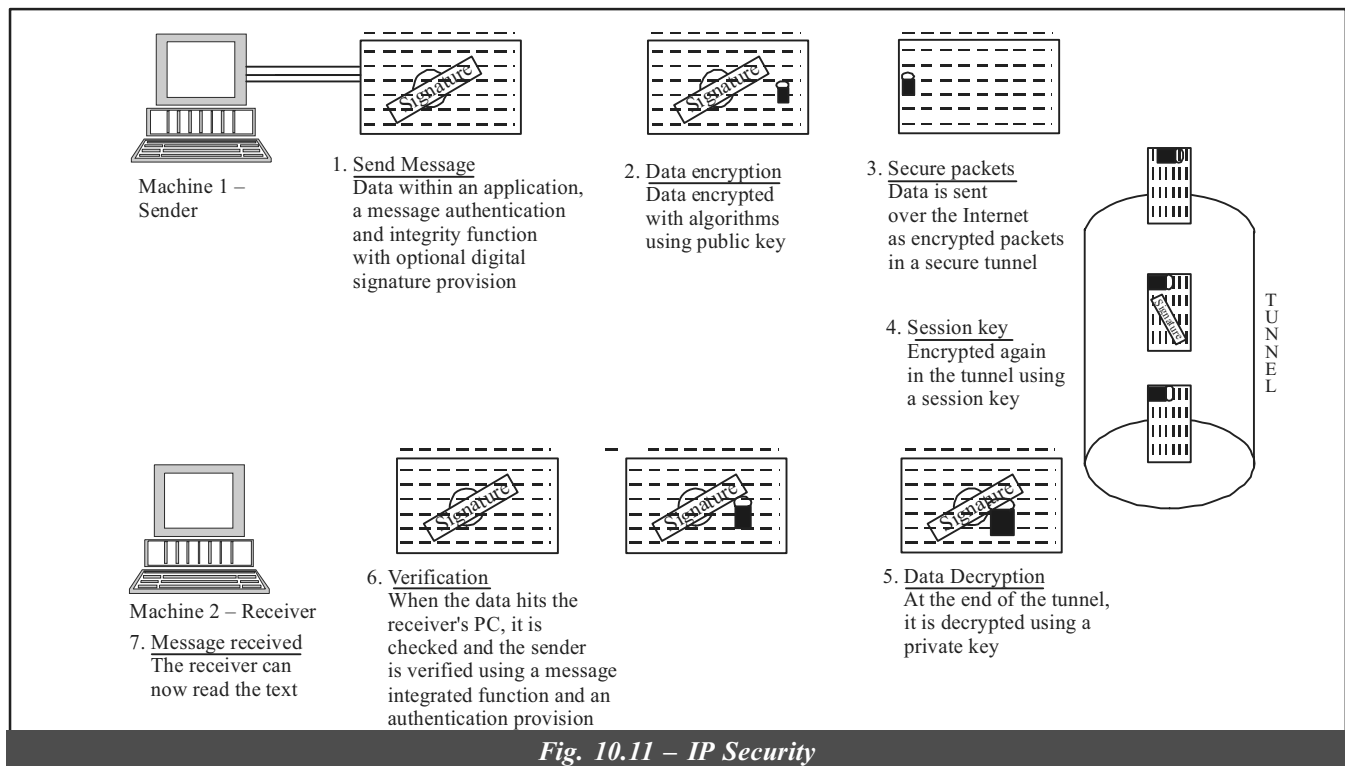


Fig. 10.11 – IP Security

10.13 TRADITIONAL PAYMENT METHODS

Specially with cheques, these are not adequate for real-time payment interaction. Real-time payments means a consumer's web browser delivers payment instructions to a merchant, who forwards those instructions to a network bank. The bank then authenticates the individuals and disburses funds to the merchant. The merchant delivers the purchased product upon customer verification.

Two fundamental assumptions are made in the case of offline payment methods.

- The transacting parties (buyers and sellers) will be in each other's physical presence (sometime).
- There will be a sufficient delay in the payment process for detection of fraud, overdraft and other problems to be identified and corrected. These assumptions are not valid on electronic commerce.

10.14 PROBLEMS WITH TRADITIONAL PAYMENT METHODS

The traditional payment methods do not work online. The reasons are due to lack of:

- Convenience
- Coverage
- Security
- Eligibility and
- Support for microtransactions.

10.15 NET BANKING

The Internet has the potential to transform the entire economics and structure of banking and personal finance so much so that traditional banking may soon become unviable from the standpoints of cost, time and convenience. And these changes, says R.G. Bhatnagar, come not so much from the customers as from the banks themselves.

It is hoped that with the advances in technology, the regulators would be able to ensure the safety of the Internet for financial transactions.

The growth of e-mail, which is almost instantaneous with no limitation of timing or location and enables one to stay globally connected, is also making a qualitative difference to retail banking. It is estimated that the data traffic via the Internet would soon outstrip voice traffic. A large number of service providers offer 'any where, any time, e-mail banks' offering service via e-mail will have a headstart over their competitors.

Advances in telecommunications are also having an influence on the ways banking would be conducted in future. A new system to make conference calls more flexible and cost-effectively has been developed by a British telecommunications company.

The system called 'Meeting Point Financial' allows banking and financial institutions to set up a conference call with branches anywhere in the world in a matter of minutes, without the expense or inconvenience of booking through an external bureau. These developments are sure to revolutionise inter-bank communication, flow of timely information and policy guidelines from the top and flow of feedback from the grassroots.

The ultimate result would be a new breed of better informed and motivated bank executives suitably equipped to deliver results. But electronic banking also carries with it some risks which are compounded by the absence of appropriate legal provisions governing internet banking and the lack of legal clarifications.

There are, in fact, many difficult issues which have to be tackled before the potential of the new technology can be exploited fully. A bank is exposed to the risk of punishment even if it unintentionally fails to comply with the existing law(s) or when the application of established rules is uncertain. As most transactions in Internet banking would be first of their kind, the applicability or otherwise of some of the provisions may require interpretation from the appropriate authority.

It is time the Reserve Bank of India (RBI) examined the Banking Companies Act and other provisions relating to banking transactions to decide how these would require modification or elaboration in the context of banking transactions in cyberspace. An authority may also have to be designated to provide advance rulings or interpretations of the existing legal provisions as far as these relate to Internet banking.

The risk of frauds and forgery is also inherent in banking transactions. It is bound to be much more in cyberspace deals, particularly because of the danger that they may not originate from the real customers. If this happens, the banks would have to bear the costs associated with such transactions and compensate parties who may suffer losses because of inadequate systems and lack of safety measures and safeguards.

As e-banking is not limited by geographical and political boundaries, it is unavoidable that specific and strict protection measures are introduced as early as possible. In the absence of appropriate measures, the customers' money can just disappear into cyberspace.

The National Consumer League in Washington recently published a list of a few most prevalent Internet frauds for the benefit of consumers and web surfers. The banks come across a lot of information about their customers and their financial position during the course of their normal business.

The banks have a legal obligation to maintain strict secrecy about the customers' affairs and not divulge this information without authorisation or due process of law. As the Internet is a public network with a free flow of information, it may not be very difficult to access information on a rival's business transactions with a bank.

Even if the secrecy promised by use of password is observed strictly, a third party may observe the information, while in transit – it could even happen by mistake. This may create complications in banks' relations with their customers and may give rise to unnecessary litigation.

In India, the lack of necessary infrastructure and the absence of effective laws make banks much more vulnerable in cyber crimes. A number of issues warranting serious consideration has surfaced in the present context of the Internet. One such concern relates to "phreaking" and "hacking".

Scammers and unscrupulous elements have always tried to take advantage of advances in technology to serve their own ends, and Internet banking would be no exception. It, in fact, provides them with a new and potentially easier mode to take unsuspecting people for a ride. The authorities concerned would have to devise ways and means to tackle hi-tech cyber crime.

A question that is agitating bankers and planners alike in the context of the Internet banking is the impact of the technology on the branch networks. Would it render branch banking redundant? Would it make the branch networks a liability? Today, size and network are important parameters of the strength of a commercial bank. For example, the State Bank of India is considered the strongest Indian Bank mainly because of its size and massive networks.

If the Internet becomes, as it promises, the primary distribution channel for the services and financial products of a bank, what will happen to the large number of branches of the commercial banks – and, more important, to the thousands of their employees – which are likely to be rendered surplus.

Though the rural branches may not be able to afford Internet banking for a long while and may therefore escape the impact, the problem is nevertheless daunting. The issue – particularly the question of surplus staff – in the context of computerisation has been debated earlier, and a number of options such as a blanket ban on further recruitments, an exit policy and a VRS (voluntary retirement schemes) have been discussed, but a viable and acceptable solution to this vexed problem has remained elusive.

Given the strong political clout of the unions and the generally pro-labour policies, a solution to the problem appears difficult, if not impossible. In brief, the latest technology holds promise of exciting changes in the financial sector. As the Internet and e-mail facilities grow, we can truly look forward to banking of international standards. This would, however, necessitate a proactive role of banks with an inclination and capacity for fairly heavy investment in technology, suitable orientation of their staff at different levels, and evolving appropriate systems and procedures to take care of the demands of new technology.

The regulators, on their part, would also have to play an equally dynamic role to ensure that proper accounting systems and disclosure norms are in place to make transactions transparent. It has, thus, to be a truly collaborative response to the challenge of technology-driven global banking involving managements, employees and the regulators. Notwithstanding the many hiccups and hurdles, the bright side is that the Internet has really caught the imagination of the youth both within and outside the banking industry, and it is sure to be an active participant in the new banking millennium – banking with a difference.

“A simple way to do that is to have a little beacon sitting in each room that says ‘I am in this room’,” Fagg said. It squirts that out of every 30 seconds, essentially using the same technology as your TV remote.” But does anybody really want a gadget that will monitor all their moves and inject its own two cents’ worth every time it wants to pop off? The key, Fagg said, is to make the system convenient but not so intrusive that it is annoying.

“What you want is a device that can assess how interruptible you are,” he said, “Maybe it can even take into account who’s calling and what they want to talk about and then make a decision as to whether or not you should be interrupted. That’s going to be critical for people accepting this as a daily device.” Fagg insists the monitor itself is relatively unobtrusive.

“You don’t really lose your field of view,” he said, because the monitor is so small you can still see the background. “You just lose binocularity in about a 10-degree wedge,” Still, it will take a bit of getting used to. The first time he wore his contraption to a faculty meeting, he generated a little anxiety among his colleagues.

“There was a lot of concern that I was videotaping everything that was going on inside this confidential faculty meeting,” he said.

E-banking – Issues and Challenges

Internet Banking is a new concept in most regions and as expected fraught with numerous challenges both on the technology and business side.

Technology Challenges

The Internet is a public network and financial transaction that is sent on this network could be trapped or altered at numerous locations. Security standards such as SSL and the public key infrastructure have emerged as the defaults to address this issue.

For enabling either information access or transactions, banks have to put up a website. There is data to suggest that most hacking that happens is targeted at breaking into website security. Typically, hackers break into a website first and then try to get access to internal systems from there. This is perhaps the most serious security threat for all banks, which need to be followed in order that the website is secure from all such threats. No security solution can offer a panacea for all known threats, but they do notify a large part of the risk involved.

Business Challenges

The transaction costs of the Internet have been stated to be the lowest of all channels. However, the total volume of benefits to the bank will become significant only when a large number of transactions are routed over the Web than over other traditional channels. This requires that the bank have a large chunk of Internet-savvy customers who can exploit this medium to its full. The Internet and the interactivity have caught the attention of many a layman to the extent that in many places, the need of the Internet itself is driving the sales of PCs up. It is a matter of time before a large chunk of the user population becomes ‘Net-savvy’.

At times, awareness levels about Internet banking remain low even to promotion of the services to the customer. As required for any other product launch, banks should have a marketing and promotion plan clearly chalked out. Also, banks can look at branding this service and creating internal organisational structure and processes to manage this brand.

Most banks are content with launching just account access services. These services though are useful but are not value adding to the extent that the customers start using the Internet banking service on a regular basis. Thus, banks need to keep introducing more and more value-added services. Ultimately, the bank's website should become a portal for all financial needs of its customers.

Banks need to look at all customer segments, apart from just retail customers, e.g., corporations or businesses and create value for them too. In fact, in some regions, it is the businesses that have ready access to the Internet. Thus, in such a scenario, creating a compelling interaction for them might make better channels including the Internet, the one-to-one personal interaction with the customers is likely to reduce. Personalised attention and service has been the cornerstone for many banks for decades and thus, losing this might ultimately mean losing customers. The bank needs to look at ways of personalising the experience of each user even on the Internet.

Internet Banking does raise quite a few issues, some of which may be quite challenging for banks. The biggest amongst perhaps is that the traditional business paradigms are demolishing much in the same way as a brick-and-mortar 'structure' would. In this regard, the business focus, operating scale and existing systems of the bank might require a close look.

However, a little effort can create innumerable new strategies, relationships, customers and 'structures'.

10.16 ENABLING REGULATIONS

Amendments to banking regulations in recent years have now enabled financial institutions in many countries to move more rapidly to adopt various forms of electronic cheque presentment and cheque truncation. Hong Kong, Australia, the United Kingdom and South Africa, for example, now allow electronic cheque clearing for value but the bank of deposit must still send the paper cheques to the paying bank eventually. In the United States, the Federal Reserve Bank is strongly encouraging industry efforts to define rules for and move toward cheque truncation, as witnessed by its strong support for PACES project.

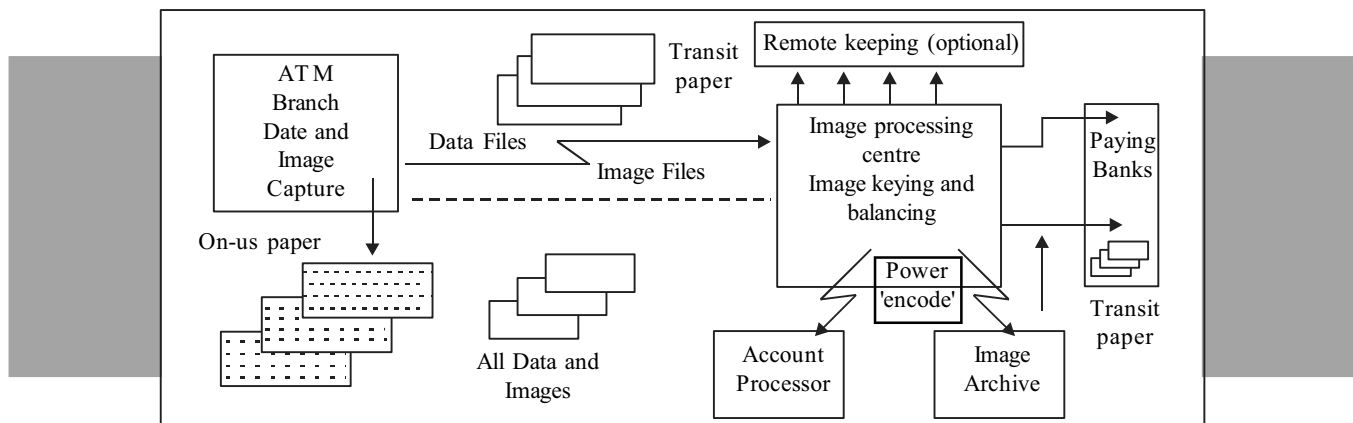


Fig. 10.12 – Branch-based Image Capture with Centralised Balancing

Singapore is set to be first country in the world to introduce nation-wide a new technology called cheque truncation. It will be implemented across the nation by 2001, said Deputy Prime Minister and Chairman of the Monetary Authority of Singapore (MAS), BG Lee Hsien Loong on 26th November, 1999.

- (i) With full cheque transaction, cheques are scanned electronically at the point of deposit, such as bank branches or client's offices.
- (ii) The digital image rather than the physical cheque issues is then transmitted to the clearing house for sorting and onward to the paying banks.
- (iii) The images of the accepted cheques are saved in a central archive while the rejected ones are re-diverted to the loop.

The technology comes from NCR, which recently showcased it at a seminar for bankers and monetary authorities for the region.

The promise – Banks (in Singapore) can save Singaporean dollar 46 million a year, according to the Monetary Authority of Singapore Estimates.

10.17 CRITICAL BUILDING BLOCK FOR ELECTRONIC COMMERCE

Currently, in India, the Internet is still largely used for e-mail and seeking information. A few e-retailers have set up shops on the Net, but the transaction volumes are nothing to write home about. For online commerce to emerge as a reality, it is imperative to establish sound security norms for facilitating online transactions and delivery mechanisms for order fulfillment.

To facilitate online transactions, validation of the involved parties and secure transmission of the financial data over the Net is critical. A dependable solution, such as a payment gateway, is required to provide the confidence and trust for affecting online financial transactions.

International trends have shown that 90% of online commerce is contributed by the B2B segment, whereas the balance is 10% from B2C commerce. The concern of secure connectivity and a payment fulfillment mechanism is crucial in both scenarios.

Transaction Security: The Impact of the Perception of Risk

Public perception on the risk of purchasing products over the Internet is growing. The success of well-publicised commerce sites such as rediff.com and indiainfo.com are doing much to accelerate the acceptance of online purchasing, but the risk apprehension of submitting confidential financial payment details over the Net remains a key concern area. Given the relatively insecure state of many e-commerce processes, the need to ensure that fraud becomes an unlikely event and that public confidence is encouraged to grow unabated is critical to the benefit of consumers and merchants alike.

International Scenario

Payment gateways have become the norm in the international scenario. Almost every major site has been acquiring credit cards through a payment gateway. These gateways have provided the basic infrastructure needed for secure and viable online business. The international market is dominated by a few major players like CyberCash and Netsol apart from a lot of relatively smaller players. The lessons learnt in these markets have to be emulated in the Indian scenario. With this perspective in mind, ICICI launched India's first payment gateway Payseal.

Payseal – ICICI's Payment Gateway

ICICI has launched Payseal – its payment gateway solution. The gateway has been developed along with Compaq and QSI. Financial Software and Systems is implementing the project in India. The acquiring is being done by ICICI Bank.

The working and need for Payseal will be elaborated by explaining the online and offline purchasing process.

10.18 THE OFFLINE AND ONLINE PURCHASING PROCESSES

To simplify our understanding, let us begin with business-to-consumer credit card transactions.

The customer presents his card at the merchant's shop. The merchant uses a Point-of-Sale (PoS) device to send the card information over a dial-up connection to his acquiring bank.

The acquiring bank then sends this information to the issuing bank through the card network. The issuing bank then authorises or rejects the transaction and sends the message back to the merchant over the path.

We must notice that the entire process takes place over a secure link. The customer gives his card information to the merchant in person. So, the merchant is physically present and known to the customer. The merchant transmits the information to the acquiring bank over a secure link.

The Online Purchasing Process

In the online marketplace, buyers and sellers are often unknown to each other. The process of purchasing a product electronically is very similar to the process of purchasing a product with a credit card by telephone. Of course, in this environment, the customer cannot validate the card usage with his signature, which is the norm in offline transactions. The authentication of the card in the absence of the signature is a stumbling block in the online mode.

When the customer's online shopping session is complete, the customer clicks on a hyperlink which takes him or her to the checkout page. At this stage, the customer is presented with a list of the goods marked for purchase, the total cost, shipping, handling, tax, etc. The customer can then add shipping instructions, name, address and so on.

The customer is normally given a range of payment options and some of the more common are discussed in more detail below. The most common is to use a credit card and the customer enters the card number, name on the card and expiry date. Then he simply types his credit card number into the payment form and waits for the products to be shipped to him. The only thing that needs to pass between the merchants and the buyer is the credit card number. The problem is that it's not that simple.

The online purchasing process is very similar to a real-world credit card transaction, except for two important differences:

- The information travels over a public network on the Internet. This makes the system vulnerable to unauthorised access.
- The merchant and the customer are not known to each other. The customer or the merchants may be fraudulent.

In such a case, it becomes important to have a secured way of making payments. A secured payment system would protect all the parties – the customer, the merchants and the banks. Thus arises the need for a payment gateway. The payment gateway ensures the security of the concerned parties, and thus facilitate Internet commerce.

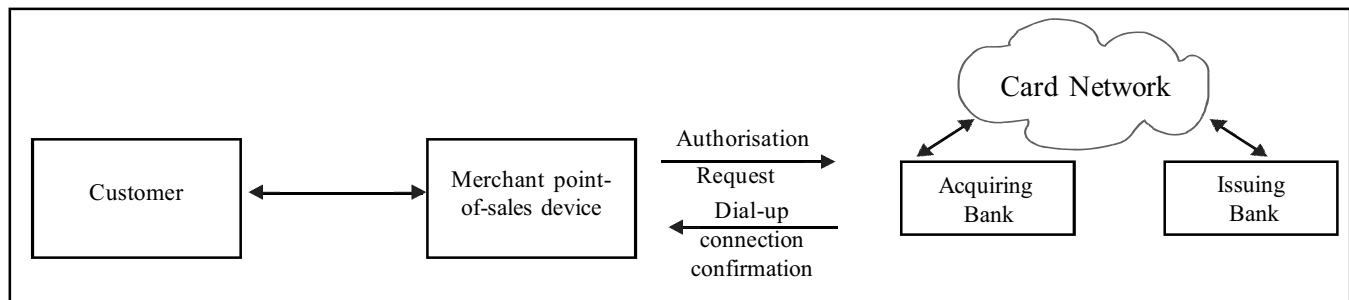


Fig. 10.13 – The Online Purchasing Process

The Solution

Customers have some legitimate fears about giving their credit card number out over the Internet. It is an open network without any basic security provisions built in. It is therefore essential that traffic be scrambled (or encrypted) and the standard SSL protocol developed by Netscape provides a high level of protection.

Unless a secure server is involved, data passes between the browser and the server unencrypted. Because of these fears, payment gateways make purchasing products online more secure. Furthermore, the customer is also wary of leaving his credit card details open to unscrupulous access at the merchant's site.

It is important for customer confidence that the site switches to secure mode as soon as credit card information is requested.

The technology used is called SSL (Secure Socket Layer). This means that all communications with the server are encrypted in such a way that eavesdroppers cannot steal the credit card information.

Customers rightly emphasise the need for customers only to give up confidential information in a secure session and users will look out for it.

10.19 THE SHOPPING PROCESS WITH PAYSEAL

In a typical retail purchase on the Internet:

- The customer fills his shopping form on a merchant's website and proceeds to check out.
- The transaction information is transmitted to the merchant's server.
- The web merchant forwards a digital order to the Payseal server in an encrypted format.
- Payseal authenticates the merchant and provides a payment option and payment details screen directly on the customer's browser over a secure 128-bit SSL + connection.
- The customer provides his credit card details, which is directly sent to the payment server.

- The credit card details are then switched to ICICI Bank for authentication. ICICI Bank then transmits the message to the card holder's (issuing) bank for payment authorisation. The issuing bank authorises the payment and transmits the confirmation bank to the payments gateway through the acquiring bank.
- On receiving authentication and authorisation, Payseal forwards the validation of the payment instrument to the merchant server.
- The merchant transmits the acknowledgment of the payment to the customer's browser.

The entire process integrates seamlessly with the shop and buy application of the web merchant ensuring a pleasant shopping experience for the customer.

The credit card details of the customer remain unknown to the Net merchant.

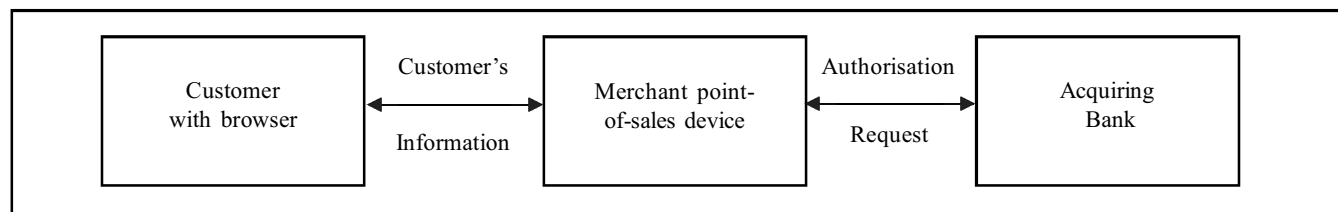


Fig. 10.14 – The Shopping Process

The Security Proposition

Even if the customer is protected by SSL technology, it is clearly important that the information remains secure. Payseal's security measures make sure that the customer's credit card details are kept secure while transacting on the Net, preventing unauthorised access. The card details would not be disclosed even to the merchant. Payseal adopts the SSL (Secure Socket Layer) technology, an internationally proven and widely accepted technology.

- Payseal is an integrated solution employing 128-bit encryption to ensure security of online transactions. In fact, ICICI Bank was the first bank to offer 128-bit security in India.
- Payseal uses a stronghold web server.
- The client's software installed at the client's end, encrypts transaction information using 280-bit RSA before passing it through an SSL pipe using 128-bit encryption.
- The data is stored at ICICI's own state-of-the-art data centre, secured by sophisticated firewalls and other network security infrastructure. Physical access to the data centre is restricted by use of biometrics.
- The gateway employs both software and hardware encryption to ensure the highest level of security.
- Payseal is protected by extensive firewalls. The gateway server is assigned a server ID and authenticated by "Verisign", one of the leading certification authorities. It may be noted that Payseal can be adapted to support SET.

Payseal Features

Payseal provides a set of unique features, which ensure fast and easy integration to your systems and also maintenance of your records.

Pre-registration and Enablement (PRE)

Delivery of payment capability to websites is accomplished through PRE, a web-based registration engine which configures software for the specific merchant environment. This allows mass registration and payment enablement with a minimum of human intervention. It totally eliminates integration, which costs tens of thousands of dollars today.

Templates at Gateway

Each merchant's payment options are delivered to consumers in the form of a template from the bank's gateway. This means that upgrading merchant's payment options can be done automatically and immediately. This contrasts greatly with present systems that require hands-on changes and additional integration.

Merchant Administration

Each merchant is allowed to reconcile, review individual transactions and assign different levels of authority for refunds, voids and captures.

10.20 ICICI PLAN TO CREATE A VIRTUAL BANK FOR VIRTUAL CUSTOMERS

ICICI's (Industrial Credit and Investment Corporation of India) e-commerce initiative are in three broad areas:

- Web enablement of their business.
- Incubation of ideas to enhance their customer base and
- Investments in the Internet and technology space across a broad spectrum of industries and areas.

The growth of their e-commerce initiatives has been considerable. They have successfully developed and launched several products over the last few months including cafemumbai.com, billjunction.com, traveljini.com and Payseal, the payment gateway service.

They have incubated over eight ventures in the last six months and are actively pursuing other areas. The driving forces have been the vision and commitment of ICICI to be a dominant player in the Internet space.

ICICI Bank launched virtual, Brand-free Banking in 2018 (offering facilities like E-wallet which offers information through a single interface to facilitate banking transactions over the Internet ICICI Bank has 15.4 million virtual addresses on UPI.

India is lagging behind the advanced countries in infrastructure availability. But the pace of adaptation of technology in this space has been much faster than witnessed in most other countries.

In fact, India is one of the fastest growing Internet markets in the world as indicated by the Gartner Group and other international analysts. The marked difference in the last few years has been the global perspective taken by several corporates in its offerings to customers in India and abroad.

E-commerce is revolutionising the way business and consumers transact across the globe. In India, online commerce is becoming a reality as the number of online users is expected to cross 70 million and the total value of online commerce is estimated to grow to \$42 billion by 2004, according to Goldman Sachs.

According to ICICI total understanding and ownership of the consumer is the key factor for success. This would require keeping abreast of the latest trends in technology and constantly upgrading the offering to corporates and consumers alike. The cornerstones for development of e-commerce are reliable delivery systems and transaction security. Initiatives such as payment gateways are critical to tapping the potential of e-commerce in India.

Virtual company is a company that is already real, but is going backwards to 'almost becoming' a company in the modern sense of the word. A company that is almost a company or 'virtual' is moving away from having physical operations such as offices, staff or physical assets. The virtual company has a structure and it conducts commerce, but without a physical location or people. The concept of an entirely virtual corporation is an unattainable goal for most organisations and deserves the same level of respect of the paperless office. Many companies are starting to incorporate virtual concepts into their operations as part of a general movement towards e-commerce.

10.21 PAPERLESS CHEQUE PROCESSING

The emergence of paperless cheque processing solutions could help bankers and monetary authorities save millions in processing costs. This can be achieved if a full cheque transaction system is implemented, from the point of deposit to clearance, final settlement and archival.

"The banking industry today is facing an increasing volume of cheque transactions, growing competitive pressures and shrinking profit margins that makes it more urgent to process existing volumes as efficiently as possible," said Mr. Mariano S. Roldan, who heads the Paperless Cheque Exchange and Settlement (PACES) project of the Financial Services Technology Consortium (FSTC) in the US. He addressed a group of bankers and monetary authorities at a seminar organised by NCR in Singapore on April 11, 2000, where NCR unveiled the world's first full cheque truncation solution to help bankers in the region enjoy the benefits of paperless cheque processing in the world of e-commerce.

While the regulatory framework required for full truncation among all financial institutions is still being developed in most countries, work is well under way on standardised, vendor neutral platforms that will enable all institutions throughout a country to enjoy the benefits of electronic cheque presentment and cheque image exchange, with truncation of paper cheques at the bank of first deposit.

Following is a pictorial representation of one of the possible scenarios of how cheque truncation can be achieved using the **NCR ECP-IX Solution**. ECP-IX stands for Electronic Cheque Presentment with Image Exchange. The top diagram

depicts what happens at the Bank of First Deposit in an image-enabled environment, while the bottom diagram shows how image-based clearing and settlement will take place between the Bank of First Deposit and the Paying Bank.

Enabling Technologies

The NCR cheque truncation solution provides the banking industry with a solution that, for the first time, will allow for image-enabled electronic cheque presentment and exchange. Components of the NCR cheque truncation solution have already been proven successful in the marketplace.

For truncation to be commercially practical, financial institutions must be able to capture cheque images of sufficient quality for settlement and fraud detection and to store these images economically. Five main types of technology have combined to make these capabilities a reality:

- (i) Advanced imaging and security technology to capture cheque images and the software required to filter and compress them for economical storage and delivery.
- (ii) Image capture transport technology to handle the different price performance and volumes of cheques.
- (iii) Client-server computing to distribute image capturing, processing and delivery functions across the enterprise.
- (iv) Database and storage systems have dramatically improved their cost/performance ratios for maintaining high volume image archives and exchange images by bulk transfer on media such as CD-ROM.
- (v) High-bandwidth telecommunications have now made it practical for financial institutions to exchange high volumes of images via broadband networks.

It is understood from the lead NCR head in India that paper cheques are handled at least 10 times and needs at least 24 hours to be cleared. 'Even the local ports only take 6 hours to clear a twenty ton container'.

10.22 E-CASH

The cost of each transaction using credit cards/cheques is quite large and it is thus not appropriate for low value transactions. Cash is used for many transactions, particularly, low-value transactions. The currency system evolved over centuries and is time-tested. The major advantages of cash are:

<i>Advantages</i>	<i>Disadvantages</i>
It is guaranteed by the government and does not normally lose its value in the short range (except when there is hyper-inflation). It is universally recognised as having value and accepted as legal tender. It can be carried around. Any person having cash can exchange it for goods or services without the help of a third party such as a bank. It is anonymous. Traders cannot normally say who gave a particular currency note. Privacy of transactions is ensured because of anonymity.	It is not safe. If you lose your purse, you will be lucky to get your cash back. It is bulky. Governments do not normally print large denomination currency notes to prevent criminals from stealthily transporting large amounts of cash. Government of India only recently started printing ₹ 1,000/- currency notes. Ten years ago, the largest denomination was only ₹ 100/- to reduce black money transactions. The purpose of electronic cash. It is to mimic cash transactions with all its advantages without its disadvantages. The major problems: • Who will issue e-cash? If it is a private agency like a bank, who will guarantee the safety of one's e-cash? Deposits in scheduled banks are normally insured by Reserve Bank (upto a specific limit). • Will anonymity of e-cash transactions be ensured? • Should it be ensured? • Will e-cash issued by a private agency be universally acceptable? Can two individuals exchange e-cash without the issuing party entering into a transaction? (This is a major advantage of cash. I can borrow a hundred rupees from a friend without the Reserve Bank knowing it!). • How will it be possible to detect forgery? A person who 'buys' e-cash should not be able to spend the same cash again. Once it is exchanged for a service, it must lose its value or be taken out of circulation. • How is the cost of handling e-cash recovered? In the case of physical cash, not only was cost recovered but a profit made by what is known as seigniorage.

Table 10.2 – Advantages/Disadvantages of E-Cash

Cash transactions ensure anonymity. It is the mode of payment to extortionists, corrupt officials and tax evaders. Larger amounts of cash are bulky and difficult to transport, and hide particularly because governments do not print high denomination currency notes. Currency is normally traceable with their serial numbers. Large amounts of cash are difficult to transport across national boundaries as customs officers keep a close watch. Ultimately, the cash is deposited in a bank (for safety and to earn interest) and cash flow chokes. Except for very few countries, banks are expected to provide information about deposits to law enforcement officials to prevent illegal transactions such as trade in narcotics. Electronic cash can cross national boundaries at lightening speed. Anonymous e-cash will be difficult to trace. If there is no limit set on e-cash transaction amount, large amounts can flow with ease and law enforcement officials will be unable to trace the money. Thus, governments may ultimately set a limit (for example, ₹ 5,000) on individual e-cash transaction.

Seigniorage

The term seigniorage evolved from early history of money when kings (or seignior – a feudal lord) had the exclusive right to mint money. The difference between the cost of minting money and its face value is profit earned by the king and is called seigniorage. In modern economy, the printing cost of currency notes is not very significant. Further, the requirement which once existed of having gold reserves equivalent to currency in circulation has been dispensed with. Thus, the concept of seigniorage in a modern monetary system managed by governments is incongruous. This concept, however, becomes relevant in e-cash. E-cash can be sold to customers at face value. The interest which accrues between the time e-cash is sold and the time at which it is used (or encashed) and debited to the seller is the equivalent of seigniorage earned by an issuing bank. Unless there is enough e-cash ‘minted’ by a bank and kept in circulation, this amount may not make up for the expenses incurred by the bank in servicing e-cash. (Debit cards work on a similar idea to recover cost.)

Currently, the technology of cash transactions in e-commerce has not matured. Many competing systems are being tried out, each with its own advantages and disadvantages. Governments do not want to allow large cash transactions to take place electronically.

E-Cash Transactions

A simple method has been described for e-cash transactions. It is being used by some banks in the United States and Europe. No such system exists in India as of now. It is primarily intended for small cash transactions. The procedure is as shown in Figure 10.15.

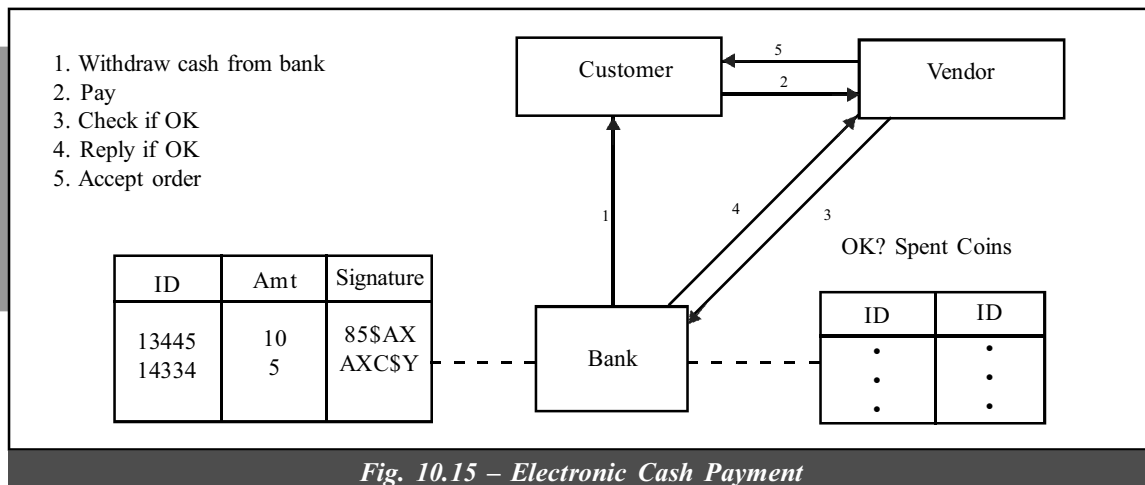


Fig. 10.15 – Electronic Cash Payment

- Step 1: A customer withdraws ‘cash’ in various denominations from the issuing bank (or financial institution) and stores it in his PC. The withdrawal takes place by the customer giving a unique identification number and denomination of each coin and requesting the bank to digitally sign it. The bank signs a coin by encrypting (<id#, denomination> with its private key. The signed e-coins are of the form <id#, denomination, bank’s signature>.
- Step 2: The customer pays a vendor for goods ordered using the signed e-coins.
- Step 3: The vendor sends the e-coin to the issuing bank for authorisation.
- Step 4: The bank checks whether the e-coin is signed by it and whether it has not been already spent. If it is a valid e-coin, it OKs the transaction and credits the amount to the vendor’s account. It puts the e-coin details in a spent e-coin database so that if the e-coin is presented again, it can dishonour it.

The communications between customer, vendor and the bank are also encrypted as the internet is used. As the amounts involved are small, symmetric cryptography is used for these communications as it is faster. There are two points, which need clarification. The first is the cost of servicing e-coins. Normally, banks will charge a small commission for the service from vendors. Second is whether a vendor who receives an e-coin from a customer can use it to purchase goods from another vendor. This is not possible as the issuing bank has to authenticate the e-coin and while doing it, it marks the coin as 'spent'. Thus, it is not really like good old cash! To mimic real cash, e-cash should have features shown in Figure 10.16.

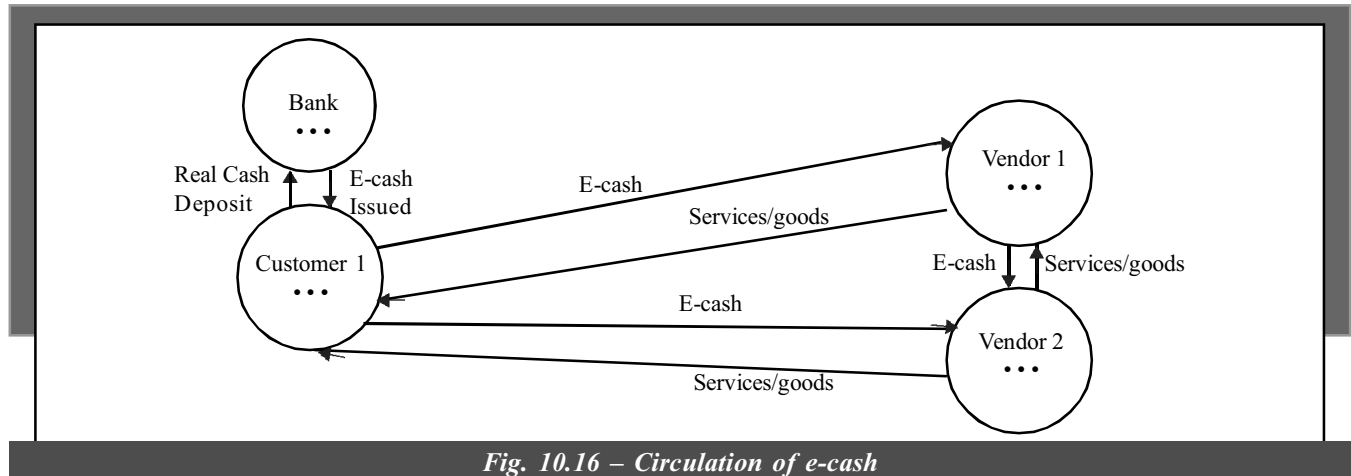


Fig. 10.16 – Circulation of e-cash

The sample protocol used above does not preserve the anonymity of cash. The bank will know which customer and vendor are involved in the cash transaction and can link the two. There is another protocol called transaction blinding in which it is possible for a customer to get e-coins issued by a bank without revealing his identity. The protocol called Chaum's blinding protocol is complicated, and as of now, not used widely.

Micropayment for Information Goods

Micro-payment is a small payment of fractions of a rupee or a dollar when 'information goods' are delivered via the internet. By information goods, we mean music files, video entertainment or text files (for example, works of fiction, software, technical information, etc.). Such a payment system is not yet in common use in India. It is being experimentally used in the United States and Europe. There are several competing systems. One such system called NetBill implemented by Carnegie Mellon University, Pittsburg, USA, in cooperation with Mellon Bank Corporation, Pittsburg, is described.

The main features of this payment system are:

1. The customer is charged only after the information is delivered.
2. The vendor is guaranteed payment when the information is delivered.
3. The customer deposits an amount which is debited when information is delivered to him. The payment system has nine steps described below (See also Figure 10.17)

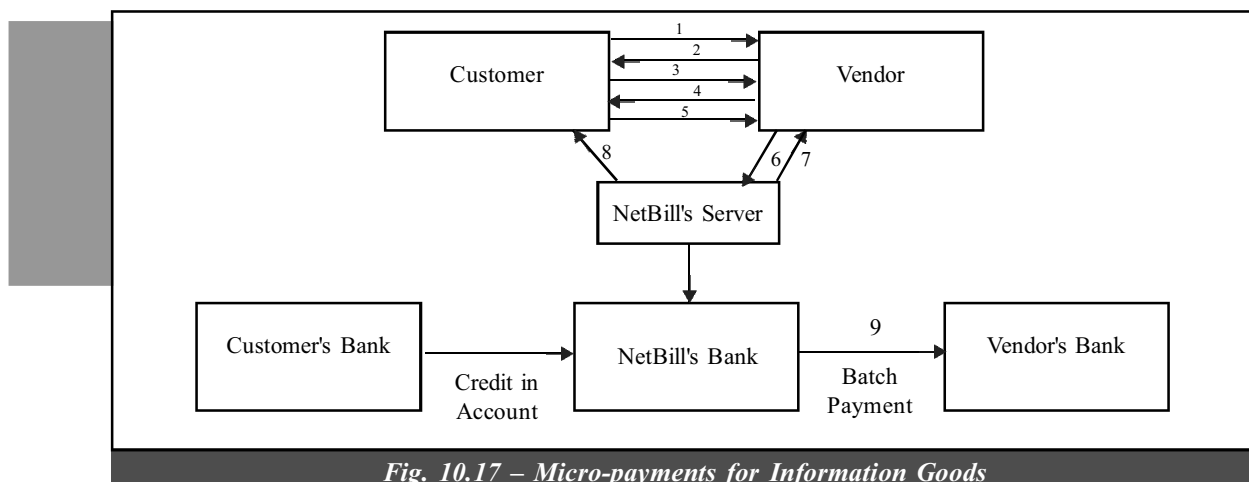


Fig. 10.17 – Micro-payments for Information Goods

- Step 1: A customer requests a quote for 'information goods' from a vendor.
- Step 2: Vendor sends a quote to the customer.
- Step 3: The customer accepts the quote and notifies the vendor of acceptance.
- Step 4: The vendor encrypts the information and sends it to the customer (note that the customer will require the key to decrypt the information).
- Step 5: The customer sends a digitally signed payment order to the vendor consisting of the checksum of encrypted information received and the amount to be paid.
- Step 6: The vendor verifies the payment order. If OK, the key to decrypt the information, along with the payment order, is digitally signed and sent to the NetBill's server by the vendor.
- Step 7: The NetBill's server checks the customer's credit balance. If it is sufficient, it debits his account based on the payment order. It credits the vendor's suspense account and sends an OK to the vendor.
- Step 8: The NetBill's server sends the decryption key to the customer to enable him to decrypt the information received from the vendor.
- Step 9: When the amount due to the vendor is substantial it is credited to his account by NetBill.

Observe that in Step 4, the customer receives the information requested in encrypted form. It can be used by him only after payment is credited to vendor's account as the decryption key is sent only in Step 8. The checksum of information is deposited with NetBill, and in case of any dispute, this provides the evidence of what accrual information was delivered. Observe also that payments are consolidated and made to the vendor periodically. This is meant to reduce the transaction cost. The system is financed by charging a commission from the vendor.

We reiterate that electronic cash payments have not yet become mature. A number of issues need resolution before a universally accepted standard is evolved.

Cyber Squatter

As Net-based shopping become popular, the value of internet addresses has been realised by both old companies and new entrants. There used to be disputes between organisations having similarity in addresses. In the initial days of the web, some enterprising individuals, who may be called as opportunists were quick to spot the profit potential in registering domain names of all kinds – many of the well-known brands – with the sole idea of selling them later for huge sums of money to firms, which wanted these names. Such organisation is known as 'cyber squatting' or 'cyber piracy'. The World Intellectual Property Organisation (WIPO) arbitration and medication center administers disputes online related to domain name registration of websites, and settles them within 45 days. The Internet Corporation for Assigned Names and Numbers known as ICANN has adopted a uniform domain name dispute resolution policy in October 1999. This policy states that an owner of a domain name being disputed by a complainant must obey the decision of an international cyber space tribunal if accused of 'bad faith' registration of domain name termed as 'cyber squatting'. If the bad faith is proved, ICANN may take stringent action to remove the domain name of the squatter and assign it to the legitimate party. However, in spite of this international policy mutually agreed, certain countries, specially USA pass their own legislation which may lead to lengthy litigation, and in some cases, severe penalties. In view of this, there is a move now to refer international disputes to WIPO which resolves disputes on copy rights, trade marks, etc. The Geneva-based UN arbitration agency was established in December 1999.

10.23 WALLET HAS MADE EASY FOR CONSUMERS

Consumers are rapidly switching nowadays to different modes of online payment that provide simple and convenient ways to transfer money from one end to the other and besides ringing in a lot of freebies.

When you go for a dinner, you may have to pay for the dinner charges even though the invitee is prepared to pay, it is a courtesy for you to accept the payment on your own charges and you do not have hard cash. At that time, in at the highend restaurant, you breeze with your smartphone and use the quick response code (QR) enabled Unified Payments Interface (UPI) app to pay the bill. Then you have to do something to call a cab to reach your destination bank using your mobile wallet. Enroute, order your grocery, call for medicines, make payment of your electricity bill from your bank account using the UPI app. Similarly, many payment transactions done and no cash need not be given. While icing on the cake is through all these transactions, one can also hope getting a lot of gifts. Hence, people are welcome to the world of digital payments through mobile wallets, internet and mobile banking, POS and others, and shares millennial. Chintan Pandya has

said that Digital payments are going to stag and the best part is that they bring in a fascinating commonality between even a street hawker as well as top-level bankers, industrialists and political honchos.

Digital Boom Responsible Factors

- During demonstration drive in India, it was digital payment platforms which were a life saver for all societal strategies including those in the hinterland.
- Stupendous growth of smartphones which assure (in the near future) product purchase facilitation by tapping our card or phone to a device.
- In India enticing offers, rewards, cashbacks and discounts being offered by digital payment vendors by decision-maker. Who doesn't like getting something 'free'.

Avers Homemakers Priya Kothari "Digital payments for one epitomise the three C's: Comfort, Convenience and Carefrees".

Adoption of digital instruments has been boosted by augmentation of merchant outlets as well as penetration of UPI which provides a simple and convenient way to transfer money across bank accounts with a large part of the activity in the digital payments space being led by the non-banking sector.

Other opinions are: "Merchants are switching to digital because it attracts more customers increases sales, facilitates ease and pace of payments, eases maintenance of sales and transaction records, and nullifies the hassle of paying change to the customer.

Top five anticipatory trends in the sector are:

- UPI • QR code (with mobile phones being the preferred payment instrument) • Zero transaction fees • Thrust on cyber security via biometric authentication, Artificial Intelligence, Machine learning and pattern matching for real-time fraud prevention for digital payment transactions • Consolidation of payments space through domination of few large players.

This justifies her loyalty to digital payments "It is extremely convenient (doesn't occupy space, no risk of money getting stolen, no hassles of change, etc.). It is easy to track expenses and who else offers you such:

- Attractive discounts
- Lots of feebies including gifts, bullion, meals at fast food outlets, a massage at your favourite spa, etc.

For some, cash payment mean:

- Less risk of overspending
- No threat of security breaches
- Less chance of being charged with promotional messages and mails (as none of the data are shared with the merchants)

To wrap up, inexorable innovation, easy-to-use payment products, interoperable payment platforms and customer awareness are expected to continue the upward trajectory of digital payments.

[Respedit.Bangalore@timesgroup.com from *The Economic Times*, 17th May, 2019, Friday, Bengaluru (India)]

10.24 MISCELLANEOUS TOPICS

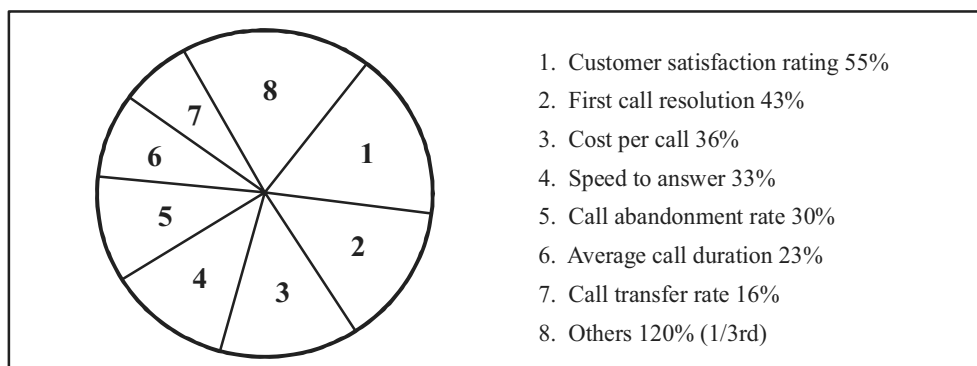


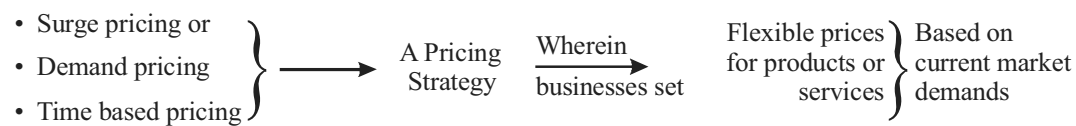
Fig. 10.18 – Customer Service Statistics

Ten of the call centre performance metrics to track:

- Average call abandoned rate
- Percentage of calls blocked
- Average time in queue
- Service level
- Average speed of answer
- Average handle time
- Average after call work time
- First call resolution

1. Dynamic Pricing

This sort of pricing is also referred to as:



Companies simply set up dynamic pricing rules and start selling more. Get a free demat support available.

2. Redefining Financial Freedom (Countrywise) (Alluva)

Adults without an account by economy (%), 2017

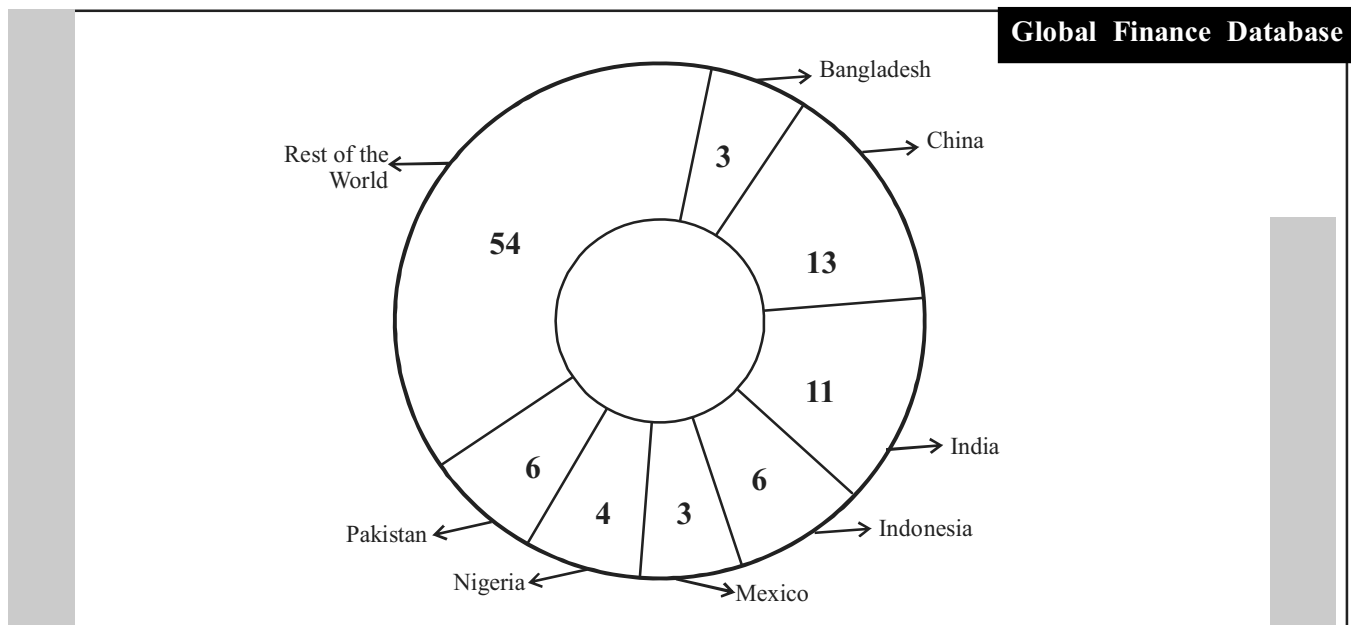


Fig. 10.19 – Adults without an Account by Economy

The World Bank Global Findex 2017 report (1.7 billion of the world approximately) do not have a bank account or any form of relationship with a financial institution. Crypto currency and block chain technology can be the catalyst for change with their potential held by these disruptive techs is being compared to the internet which revolutionised communication and information transfer. (Fundex is Financial Index)

Government and central banks around the world continue to be divided on whether Bitcoin should be treated as a commodity currency or asset.

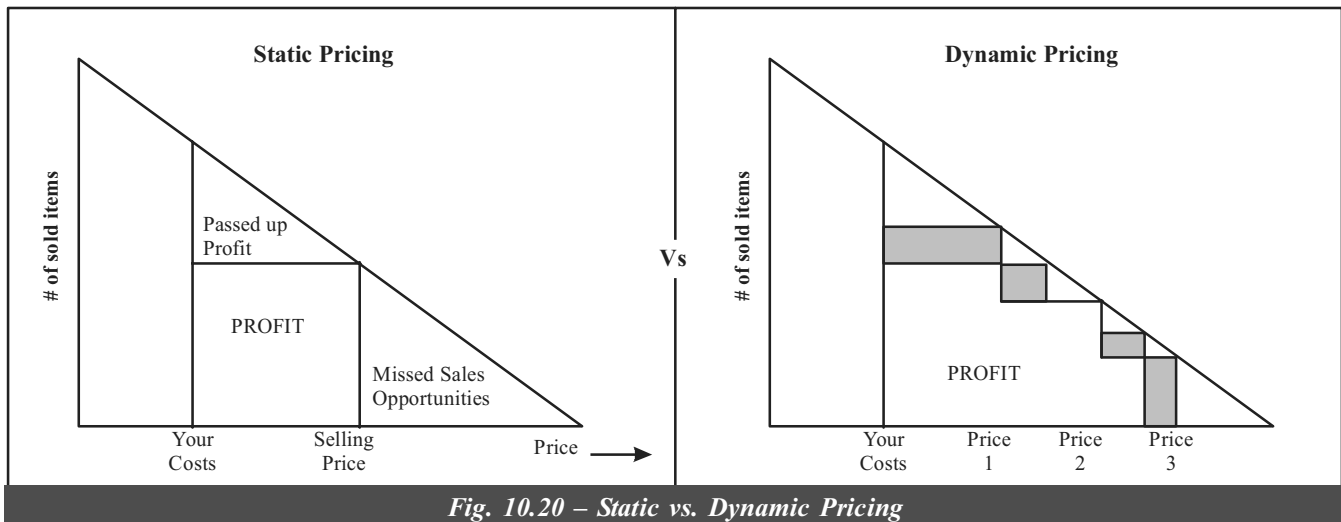


Fig. 10.20 – Static vs. Dynamic Pricing

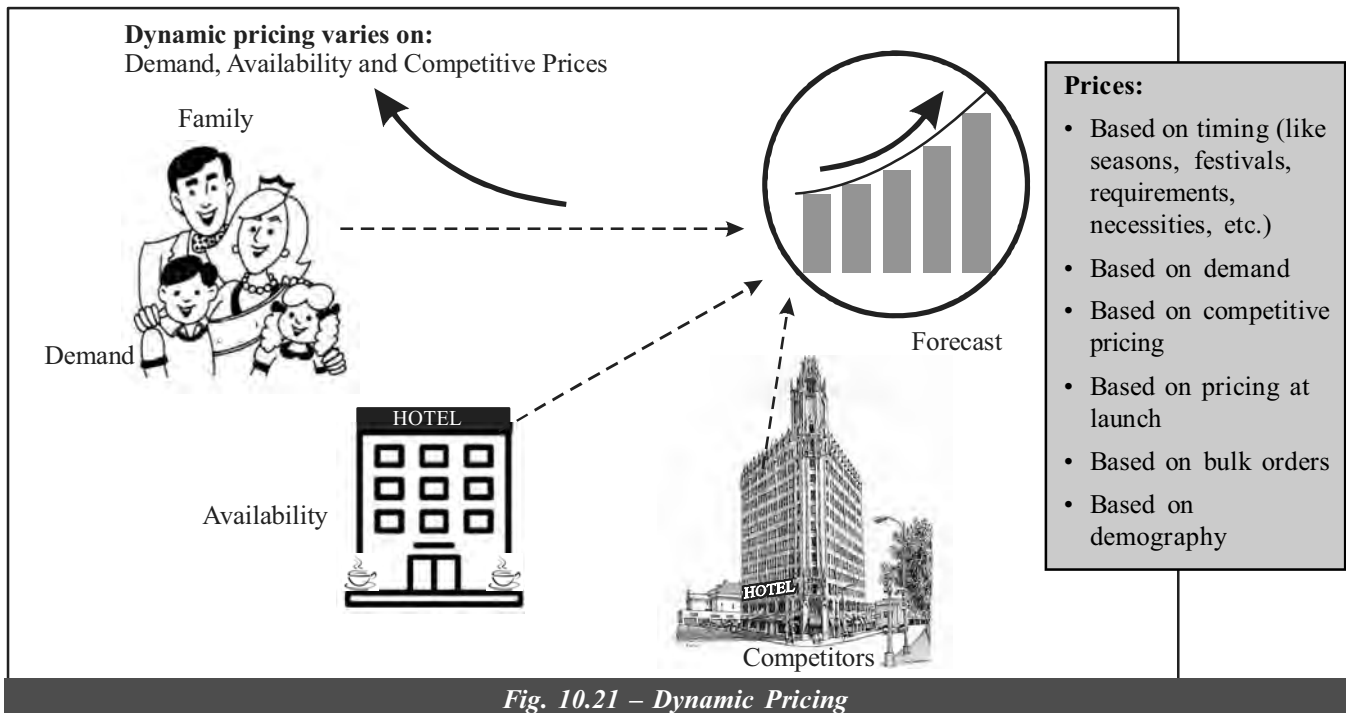


Fig. 10.21 – Dynamic Pricing

Dynamic pricing is a pricing strategy that applies variable prices instead of deciding on a set price for a season. Retailers can update their prices multiple times per day to capitalise on the every changing market. This is also “Surges pricing”, “demand pricing” or ‘time based pricing’, strategy. The same product is sold at different prices to different groups of people technically.

Value based pricing is a dynamic pricing method based in the economic principles of demand; shows the best resells in additional sales of total margin.

THE LEAST YOU NEED TO KNOW

- What is Electronic Funds Transfer? What are the different types of electronic payment systems?
- How the modern payment systems worked? An understanding of payment types and smart cards?
- What is payment security? An understanding of information sources for e-commerce and payments on the Web and Internet protocol security.
- What are B2B and B2C e-commerce payment gateways? An understanding of e-banking, issues and challenges, building blocks for e-commerce, online purchasing processes and paperless cheque processing.
- Know e-commerce payment methods, Digital payment methods, Banking cards, Online payment requirements, Electronic Bill Payments, Presentation, Online payment requirements, Static and Dynamic pricing, etc.

WRAP UP

- Electronic Funds Transfer (EFT) is an electronic transfer of information that equates to moving funds from one financial institution to another. Electronic payment systems are alternative cash or credit payment methods using various electronic techniques to pay for products and services in electronic commerce.
- With the advent of computer networking in banks, a major improvement in customer service was found. Networks enabled the concept of any branch banking. Customers can draw money from any branch. Computer networking enabled banks and their customers to overcome the tyranny of geography. To overcome the tyranny of time, ATMs were introduced.
- Two basic requirements for e-commerce are: (i) an appropriate legal framework for grants recognition to electronic records and electronic authentication of such records and (ii) availability of electronic authentication systems for transactions that are secure enough.
- Three broad categories of electronic payment systems are on the basis of Trusted third party; Variations of digital cash, electrical money and coins; and Extension of the conventional notational fund transfer.
- Modern payment systems are done through PC Banking, Credit cards, Electronic or i-cheques, Micropayments, Smart cards and e-cash.
- The techniques involved in stealing or altering the payment information by criminals are: spoofing, sniffing, content alteration, denial of service. Internet protocol security (IPSec) is a suite of protocols for secure private communications over the Internet.
- Banks understand the need to separate the content/financial product from the distribution channel (the branch). Banks recognise the potential benefits of early entry into what will inevitably be an important but crowded market. In order to avoid falling back, banks are going directly to the market with online banking programs, rather than running lengthy pilot tests with focus groups.
- If banks do not offer online banking services, affluent customers will be stolen away by software companies, online access services, brokerages, or global entertainment companies. The challenge for the banking industry lies in creating the right atmosphere and providing the right incentives for consumers to use personal computers regularly for banking, and in making sure that they provide attractive and affordable services.
- Internet banking is a new concept in most regions, and as expected, fraught with numerous challenges both on the technology and business side. The emergency of paperless cheque processing solutions could help bankers and monetary authorities to save millions in processing costs.
- The Home banking technology is also growing fast and banks are investing heavily in this sector. The four major categories of home banking are proprietary bank dial-up service, off-the-shelf home finance software, online services-based banking and worldwide web-based banking.
- Wallets have made easy for consumers to operate.

TAKING STOCK

1. What do you mean by Electronic Funds Transfer? What are the security issues involved in this?
2. What are the different types of Electronic payment systems? Explain each of them.
3. 'Research is going on into applying biometrics in the design of smart cards.' Explain.
4. 'Internet Banking is a new concept in most regions, and as expected, fraught with numerous challenges both on the technology and the business side.' Explain this in the context of e-Banking.
5. 'In India, the lack of necessary infrastructure and the absence of effective laws make banks much more vulnerable in cyber crimes.' What are your views?
6. Explain the online purchasing process. 'Online purchasing process is similar to a real-world credit card transaction except for two major differences.' Can you identify them?
7. Explain how the shopping process works with ICICI's developed software Payseal on the Internet.
8. If we say India is one of the fastest growing Internet markets in the world, do you agree? What are your views?
9. 'The Banking Industry today is facing an increasing volume of cheque transactions, growing competitive pressure and shrinking profit margins.' What is the solution for this? Has any country taken the lead to solve this problem? Which country? In what way?
10. What are your views about 'wallet have been made easy for consumers'?
11. Explain the Digital Boom responsible factors.
12. Which are the online payments requirements? How they work?
13. Explain the working of Electronic Bill Presentation and Payment (EBPP).
14. Which are the Banking Cards? Explain each one of them.
15. What do you mean by Dynamic pricing?